

PROVISION OF PERIOPERATIVE THEATRE TECHNICAL SERVICES

Learning Outcomes

1. Prepare to provide Perioperative Theatre Technical services
2. Perform Perioperative Theatre Technical services
3. Evaluate Perioperative Theatre Technical services
4. Wind-up Perioperative Theatre Technical services

Learning Outcome 1: Prepare to provide Perioperative Theatre Technical services

Introduction to the learning outcome

Upon completion of this unit describes the knowledge, skills and attitudes required to prepare for Safety practices in Perioperative Theatre, Standard operating procedures, Aseptic techniques, Perioperative Theatre layout, Perioperative Theatre instruments and apparatus, Perioperative Theatre supplies, Types of surgeries, Decontamination, Basic anaesthesia, Basic pharmacology in theatre and Perioperative Theatre ethics.

Identification and reporting of faulty Operation Theatre apparatus

Performance Standard

1. Protective gears are worn according to technical services to be offered
2. Perioperative Theatre environment is prepared in accordance with standard operating procedure
3. Required materials and apparatus are gathered according to the technical services to be provided
4. Condition of the required materials and equipment is checked according to their user manuals and manufacturer's instructions
5. Theatre equipment and apparatus are arranged according to work place set up and standard operating procedures
6. Identified problem of PT apparatus and equipment is reported according to workplace set up

Definition of terms

Disinfection: Disinfection describes a process that eliminates many or all pathogenic microorganisms, except bacterial spores, on inanimate objects

Sterilization: the process of making something free from bacteria or other living microorganisms.

Content to be covered

Learning outcome 1: **Prepare to provide Perioperative Theatre Technical services**

Safety practices in Perioperative Theatre

Employers should introduce new members of staff to basic health and safety laws and principles. Nurse training incorporates awareness of specific legislation such as Manual Handling Operations Regulations 1992, Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 1995, and Control of Substances Hazardous to Health 2002.

There are additional health and safety issues specific to theatres. During the perioperative period several members of a multidisciplinary team may be present, together with occasional visitors, who may possess varying degrees of knowledge, training and experience. All present have a joint responsibility, within a relatively small and restricted area and surrounded by specialist equipment and potential hazards, to work towards the successful completion of the patient's anaesthesia and surgery (Taylor and Campbell, 1999a).

Government initiatives such as the Essence of Care (DoH, 2001) have increased public expectations of health care. The detrimental effects of wound infection, both economically and with regard to patient morbidity, is well known. The role of decontamination as part of essential control measures is also well documented (Waller, 2002). In the theatre environment, the principles of decontamination and asepsis underpin many health and safety policies and protocols, with the aim of avoiding the introduction and spread of infection.

Special consideration must be given to patients under general anaesthetic, as they are unable to take care of their own safety needs or voice any concerns. Pressure area protection is a fundamental aspect of nursing care and particular attention should be paid to this in theatres as patients are incapable of moving at a time when additional risk factors are present (Hartley, 2003). Staff must ensure that visitors to theatres receive instruction in health and safety procedures.

During the total perioperative period health and safety procedures apply to all those who are in the theatre complex, including: All health care staff; Nurses and porters who collect and return patients to the ward; Administration staff; Sterile service unit staff; Domestic/housekeeping staff. Theatre preparation. To ensure efficient and safe running of the theatre list, it should be managed by a designated person, usually the most senior. There should be sufficient staff on duty, and they should all have a clear understanding of their role in the team, based on their skills and abilities (Taylor and Campbell, 1999b).

All equipment should be tested before it is needed, and care should be taken with electrical equipment so that cables do not present a trip hazard or run across areas where there is a risk of fluids pooling. The heating should be set at a level that is safe and comfortable for patients and staff (Parker, 1999).

Infection control

A meticulous theatre-cleaning regime is fundamental to preventing infection. Theatres should be clean and dust free, ventilation systems must be in working order and doors must be kept closed.

All items needed for the operation should be in place and an adequate supply of regularly used items should be on hand to minimise movement to and from theatre during the operation (Taylor and Campbell, 1999b). The theatre floor should be cleaned thoroughly every day, and floors and surfaces should be damp-dusted between operations, ensuring any spillage of blood or body fluids is removed (Gould, 2001). A cleaning agent of proven activity should be used, cloths should be disposable and mop heads should be sterilised daily (Line, 2003).

Cleaning, disinfection and sterilisation

Reusable instruments should be processed according to local policy, along with theatre drapes and reusable scrub gowns. Decontamination is defined by Waller (2002) as 'the combination of processes including cleaning, disinfection and sterilisation, used to make a reusable device safe for further use on patients and handling by staff'. Miller (2002) suggests that the terms cleansing, disinfecting and sterilisation are used synonymously when they have distinct functions .

Sterile field

This consists of the instrument trolleys, the scrub team and the draped patient. There must be a wide margin of safety marked by an invisible wall at the trolley edges, which marks the barrier over which non-sterile items should not cross.

An aseptic technique is essential and sterile items must be received from the circulating person in a manner that does not contaminate the field. Any breaks in sterility or the sterile field should be reported immediately to the scrub person so that the contaminated item can be isolated and removed if possible (Taylor and Campbell, 1999a).

Instruments should be inspected by the scrub nurse and rejected if any visible sign of contamination remains after the sterilisation process.

The spread of resistant bacteria such as methicillin-resistant *Staphylococcus aureus* (MRSA) and conditions such as variant Creutzfeldt-Jakob disease (CJD) are well publicised by the media and are of particular concern (Waller, 2002). Patients who pose a high risk of infection should be left until the end of the case list and universal precautions strictly observed.

Protective clothing

Protective clothing protects theatre staff from potential infection from pathogenic micro-organisms and prevents clothing becoming wet or soiled (Xavier, 1999). Any person who enters a theatre should wear theatre pyjamas. Line (2003) explains that poly-cotton material allows bacteria through its weave and is easily dampened. This presents a hazard to staff and patients as body fluids can be absorbed and transferred.

The scrub practitioner wears a sterile, long-sleeved gown that is laundered after each theatre case. With the introduction of new European Union rules regarding the testing of re-usable materials it has been suggested that it will be increasingly common to use single-use gowns and drapes that are designed to resist wetting, tearing and bacterial penetration and dispersal (Line, 2003).

Footwear and gloves

Outside shoes should never be worn in theatre. The hospital should supply non-slip, anti-static theatre clogs or boots, which should be washed after each use.

Surgeons and scrub nurses who come into contact with sterile body areas wear sterile gloves (often two pairs). They should be put on according to the local scrubbing-up policy,

ensuring an aseptic technique and taking care that the barrier system of gown and gloves is not broken where the two meet (Line, 2003; Parker, 1999).

Although wearing gloves does not prevent accidental needlestick injuries, the DoH (1998b) suggests that they help reduce the volume of blood to which the wearer is exposed. Clean gloves should be put on by circulating staff before touching patients or any equipment, specimens, or drapes that could be contaminated.

Theatre technicians must wash their hands after wearing gloves as bacterial growth increases in the moisture that accumulates under them (Xavier, 1999). Concern has been raised over sensitivity to latex, therefore gloves should be worn only when necessary and removed immediately after completion of the task (Perry and Barnett, 1998).

Theatre hats and masks

Theatre hats are worn routinely, and prevent loose hair and skin from falling on wounds or equipment (Parker, 1999). They are only fully effective if they cover the scalp and also the forehead if the wearer has a fringe.

The surgical mask was introduced over a century ago. Its continued use is controversial as evidence is not conclusive that wearing a mask reduces postoperative infection. Wearing a mask, however, does protect the practitioner from contact with body fluids (Parker, 1999).

When removed, care should be taken to handle masks by the strings so as not to contaminate the hands with any biological matter and they should be disposed of immediately (Perry and Barnett, 1998).

Sharps, swabs and instruments

It is recommended that counts of swabs, instruments (including scratchpads and diathermy tips), and sharps (needles and blades) should be undertaken at strategic points during an operation. This is to ensure no foreign body remains in a patient after closure of a wound (National Association of Theatre Nurses, 1998).

The scrub practitioner is responsible for counting all items used in an operation before incision, at first layer closure, and at skin closure. It is also standard practice to give a verbal report to the surgeon at the end of the procedure to the effect that all sharps, swabs, and instruments are correct (Pirie, 2004).

The count requires full concentration, should be audible, and should be carried out by two people, one of whom must be qualified. Both people should sign the theatre register.

The initial count should be recorded before the operation begins and any items that are added during the course of the operation should be recorded by the circulating nurse. All items counted during the course of the operation should be recorded clearly on a dry-wipe board. Items removed from the sterile field should be visibly deleted with a scoring line or cross (Pirie, 2004; Taylor and Campbell, 1999a).

Each swab counted must be checked to ensure its radiopaque strip is intact. This is because, in the event of a lost swab (or sharp, or instrument), the surgeon might request an X-ray to ensure that it has not been left in the patient. Any item deliberately left inside the patient should be clearly documented in the patient's notes.

Every sharp object in a theatre should be regarded as a potential source of injury. Accepted practices in the handling of sharps should be adhered to, for example avoiding manual re-sheathing of needles, and disposing of sharps in containers that comply with the British Standard 7320.

Needles, blades or instruments should never be passed directly hand to hand, but should be passed via a 'neutral zone', for example a receiver (Perry and Barnett, 1998). Needles and blades should be mounted for use using the correct instrument, never the fingers. Any sharps injury should be managed and reported to occupational health according to local policy.

Specialist equipment

Electrosurgery (diathermy)

Electrosurgical (diathermy) devices use high frequency electrical current to generate heat, which can cut and coagulate body tissue (Brown, 2000). Diathermy may be used in either monopolar or bipolar mode. The patient's skin should be evaluated before and after diathermy use. Correct use of equipment will ensure that skin integrity remains intact. The patient should not be in contact with grounded metal objects as the provision of an alternative pathway for the electrical current could result in a burn. Any metal parts of the operating table that could come into contact with the patient should therefore be padded and jewellery and piercings should be removed before arrival in theatre.

Monopolar mode

In monopolar mode the electrical current flows from the generator in the electrosurgery apparatus to an active electrode, which is the surgeon's tool. The current continues through the patient to an adhesive dispersive plate (the return electrode) that is placed appropriately on the patient's body so that it is 'earthed' safely back to the electrosurgery unit, thereby preventing an electrical burn. The return electrode should be placed over a vascular, muscular area to promote electrical conductivity and to dissipate heat.

Shaving may be necessary as hair at the dispersive site prevents complete plate contact with skin. The plate should not be placed over bony prominences (this can impede return current), over implanted prostheses (there is a potential for overheating), over scar tissue or over areas distal to tourniquets where adequate tissue perfusion cannot be guaranteed.

Bipolar mode

In bipolar mode both active and return electrodes are combined within the same instrument. One prong of the bipolar forceps is active and the other is the return electrode. This system is safer because the patient does not form part of the electrical circuit and therefore a dispersive plate is not required. If the patient has a pacemaker, this is the system of choice to avoid a monopolar current passing through the heart. However, the low power, low voltage waveform used with this system means it is usually incapable of cutting (Wicker, 2000).

X-rays

Precautions should be taken to protect theatre personnel from the damaging effects of exposure to ionising radiation, which are well understood compared with other occupational risks. In accordance with the Ionising Radiation Regulations 1999, staff who work with ionising radiation should limit exposure to no more than is reasonably necessary, and should exercise reasonable care while carrying out such work.

Every employee who works with ionising radiation should make full and proper use of the protective equipment provided. The number of people present in theatre should be limited to those necessary for the procedure and those present must wear lead aprons. It is advisable for pregnant women to avoid X-rays because radiation may cause foetal abnormalities (Smith, 2000).

Hazards from volatile liquids or gases

Theatre ventilation systems help to extract toxic gases that could lead to fatigue, nausea, headaches and may play a part in a range of health problems including renal, hepatic, and neural diseases (Pressly, 2000). Therefore any system-failure warning lights should be reported, investigated and rectified promptly.

Parker (1999) states that the minimum requirement for air changes is 20 per hour in the actual operating room and the sterile supplies room. The smoke produced when electrosurgical devices are used contains toxic substances that present biological and chemical hazards, so extraction systems should be used during procedures involving diathermy (Wicker, 2000).

It is advisable to avoid the use of flammable solutions with diathermy as there is the risk of ignition. Methyl methacrylate is a volatile flammable liquid used in orthopaedic surgery to make bone cement. The vapour given off on mixing may cause irritation of the eyes and the respiratory passages or a general feeling of ill health. Anyone affected should leave the theatre immediately. Soft contact lenses should be removed when methyl methacrylate is being mixed as they are permeable to irritant vapours.

Inhalation anaesthetics may be gases or volatile liquids and require suitable equipment for storage and administration. Gas cylinders used in theatre must be clearly labelled with the name or chemical symbol of the gas on the shoulder of the cylinder and the valve. They should be stored in a cool, well-ventilated room, free from flammable materials. No lubricant of any description should be used on the cylinder valves.

Nitrous oxide, used in anaesthesia, can be dangerous to both patient and staff if used for a prolonged period in an inadequately ventilated space. Side-effects can include megaloblastic anaemia and the depression of white cell formation (British Medical Association, 2000).

Waste disposal

Parker (1999) explains that theatre design should incorporate a sequence of clean zones from the entrance to the operating area, to allow easy movement of staff from one clean area to another.

Conversely, waste and contaminated items should be removed through 'dirty' areas without passing through clean areas. Body fluids collected in suction bottles should be treated with a gelling agent and allowed to solidify before being double-wrapped to

prevent leakage (see case study, Box 2), as should waste tissue and bone fragments. All clinical waste (placed in yellow bags) should be labelled with the date, theatre number and case number to allow traceability.

From an environmental point of view, machines are now available that use microwaves to make used materials safe for disposal in landfill sites. Normal rubbish should be placed in black bags and disposed of in landfill sites to avoid incineration and minimise air pollution.

Standard operating procedures

A Standard Operating Procedure is a document which describes the regularly recurring operations relevant to the quality of the investigation. The purpose of a SOP is to carry out the operations correctly and always in the same manner. A SOP should be available at the place where the work is done"

Aseptic techniques

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Figure 1

Definition of aseptic technique

Aseptic technique is a set of specific practices and procedures performed under carefully controlled conditions with the goal of minimizing contamination by pathogens.

Purpose

Aseptic technique is employed to maximize and maintain asepsis, the absence of pathogenic organisms, in the clinical setting. The goals of aseptic technique are to protect the patient from infection and to prevent the spread of pathogens.

Often, practices that clean (remove dirt and other impurities), sanitize (reduce the number of microorganisms to safe levels), or disinfect (remove most microorganisms but not highly resistant ones) are not sufficient to prevent infection.

Asepsis and Sterile Technique

The Centers for Disease Control and Prevention (CDC) estimates that over 27 million surgical procedures are performed in the United States each year.

Surgical site infections are the third most common nosocomial (hospital-acquired) infection and are responsible for longer hospital stays and increased costs to the patient and hospital. Aseptic technique is vital in reducing the morbidity and mortality associated with surgical infections.

Description

Aseptic technique can be applied in any clinical setting. Pathogens may introduce infection to the patient through contact with the environment, personnel, or equipment.

Asepsis in the operating room

All patients are potentially vulnerable to infection, although certain situations further increase vulnerability, such as extensive burns or immune disorders that disturb the body's natural defenses. Typical situations that call for aseptic measures include surgery and the insertion of intravenous lines, urinary catheters, and drains.

Surgery Aseptic Technique

Aseptic technique is most strictly applied in the operating room because of the direct and often extensive disruption of skin and underlying tissue. Aseptic technique helps to prevent or minimize postoperative infection.

PREOPERATIVE PRACTICES AND PROCEDURES

The most common source of pathogens that cause surgical site infections is the patient. While microorganisms normally colonize parts in or on the human body without causing disease, infection may result when this endogenous flora is introduced to tissues exposed during surgical procedures.

In order to reduce this risk, the patient is prepared or prepped by shaving hair from the surgical site; cleansing with a disinfectant containing such chemicals as iodine, alcohol, or chlorhexidine gluconate; and applying sterile drapes around the surgical site.

In all clinical settings, handwashing is an important step in asepsis. The “2002 Standards, Recommended Practices, and Guidelines” of the Association of Perioperative Registered Nurses (AORN) states that proper handwashing can be “the single most important measure to reduce the spread of microorganisms.”

In general settings, hands are to be washed when visibly soiled, before and after contact with the patient, after contact with other potential sources of microorganisms, before invasive procedures, and after removal of gloves. Proper handwashing for most clinical settings involves removal of jewelry, avoidance of clothing contact with the sink, and a minimum of 10–15 seconds of hand scrubbing with soap, warm water, and vigorous friction.

A surgical scrub is performed by members of the surgical team who will come into contact with the sterile field or sterile instruments and equipment. This procedure requires use of a long-acting, powerful, antimicrobial soap on the hands and forearms for a longer period of time than used for typical handwashing.

Institutional policy usually designates an acceptable minimum length of time required; the CDC recommends at least two to five minutes of scrubbing. Thorough drying is essential, as moist surfaces invite the presence of pathogens. Contact with the faucet or other potential contaminants should be avoided. The faucet can be turned off with a dry paper towel, or, in many cases, through use of a foot pedal.

An important principle of aseptic technique is that fluid (a potential mode of pathogen transmission) flows in the direction of gravity. With this in mind, hands are held below elbows during the surgical scrub and above elbows following the surgical scrub. Despite this careful scrub, bare hands are always considered potential sources of infection.

Sterile surgical clothing or protective devices such as gloves, face masks, goggles, and transparent eye/face shields serve as barriers against microorganisms and are donned to maintain asepsis in the operating room.

This practice includes covering facial hair, tucking hair out of sight, and removing jewelry or other dangling objects that may harbor unwanted organisms. This garb must be put on with deliberate care to avoid touching external, sterile surfaces with nonsterile objects including the skin.

This ensures that potentially contaminated items such as hands and clothing remain behind protective barriers, thus prohibiting inadvertent entry of microorganisms into sterile areas. Personnel assist the surgeon to don gloves and garb and arrange equipment to minimize the risk of contamination.

Donning sterile gloves requires specific technique so that the outer glove is not touched by the hand. A large cuff exposing the inner glove is created so that the glove may be grasped during donning.

It is essential to avoid touching nonsterile items once sterile gloves are applied; the hands may be kept interlaced to avoid inadvertent contamination. Any break in the glove or touching the glove to a nonsterile surface requires immediate removal and application of new gloves.

Asepsis in the operating room or for other invasive procedures is also maintained by creating sterile surgical fields with drapes.

sterile drapes are sterilized linens placed on the patient or around the field to delineate sterile areas. Drapes or wrapped kits of equipment are opened in such a way that the contents do not touch non-sterile items or surfaces.

Aspects of this method include opening the furthest areas of a package first, avoiding leaning over the contents, and preventing opened flaps from falling back onto contents.

Equipment and supplies also need careful attention. Medical equipment such as surgical instruments can be sterilized by chemical treatment, radiation, gas, or heat. Personnel can take steps to ensure sterility by assessing that sterile packages are dry and intact and checking sterility indicators such as dates or colored tape that changes color when sterile.

INTRAOPERATIVE PRACTICES AND PROCEDURES

In the operating room, staff have assignments so that those who have undergone surgical scrub and donning of sterile garb are positioned closer to the patient. Only scrubbed personnel are allowed into the sterile field. Arms of scrubbed staff are to remain within the field at all times, and reaching below the level of the patient or turning away from the sterile field are considered breaches in asepsis.

Other “unscrubbed” staff members are assigned to the perimeter and remain on hand to obtain supplies, acquire assistance, and facilitate communication with outside personnel. Unscrubbed personnel may relay equipment to scrubbed personnel only in a way that preserves the sterile field.

For example, an unscrubbed nurse may open a package of forceps in a sterile fashion so that he or she never touches the sterilized inside portion, the scrubbed staff, or the sterile field. The uncontaminated item may either be picked up by a scrubbed staff member or carefully placed on to the sterile field.

The environment contains potential hazards that may spread pathogens through movement, touch, or proximity. Interventions such as restricting traffic in the operating room, maintaining positive-pressure airflow (to prevent air from contaminated areas from entering the operating room), or using low-particle generating garb help to minimize environmental hazards.

Other principles that are applied to maintain asepsis in the operating room include: All items in a sterile field must be sterile.

Sterile packages or fields are opened or created as close as possible to time of actual use.

Moist areas are not considered sterile.

Contaminated items must be removed immediately from the sterile

Only areas that can be seen by the clinician are considered sterile (i.e., the back of the clinician is not sterile).

Gowns are considered sterile only in the front, from chest to waist and from the hands to slightly above the elbow.

Tables are considered sterile only at or above the level of the table.

Nonsterile items should not cross above a sterile field.

There should be no talking, laughing, coughing, or sneezing across a sterile field.

Personnel with colds should avoid working while ill or apply a double mask.

Edges of sterile areas or fields (generally the outer inch) are not considered sterile.

When in doubt about sterility, discard the potentially contaminated item and begin again.

A safe space or margin of safety is maintained between sterile and nonsterile objects and areas.

When pouring fluids, only the lip and inner cap of the pouring container is considered sterile; the pouring container should not touch the receiving container, and splashing should be avoided.

Tears in barriers and expired sterilization dates are considered breaks in sterility.

General habits that help to preserve a clean medical environment include:

safe removal of hazardous waste, i.e., prompt disposal of contaminated needles or blood-soaked bandages to containers reserved for such purposes

prompt removal of wet or soiled dressings

prevention of accumulation of bodily fluid drainage, i.e., regular checks and emptying of receptacles such as surgical drains or nasogastric suction containers

avoidance of backward drainage flow toward patient, i.e., keeping drainage tubing below patient level at all times

immediate clean-up of soiled or moist areas

labeling of all fluid containers with date, time, and timely disposal per institutional policy

maintaining seals on all fluids when not in use

The isolation unit is another clinical setting that requires a high level of attention to aseptic technique. Isolation is the use of physical separation and strict aseptic technique for a patient who either has a contagious disease or is immunocompromised. For the patient with a contagious disease, the goal of isolation is to prevent the spread of infection to others. In the case of respiratory infections (i.e., tuberculosis), the isolation room is especially designed with a negative pressure system that prevents airborne flow of pathogens outside the room.

The severely immunocompromised patient is placed in reverse isolation, where the goal is to avoid introducing any microorganisms to the patient. In these cases, attention to aseptic technique is especially important to avoid spread of infection in the hospital or injury to the patient unprotected by sufficient immune defenses.

Entry and exit from the isolation unit involves careful handwashing, use of protective barriers like gowns and gloves, and care not to introduce or remove potentially contaminated items. Institutions supply specific guidelines that direct practices for different types of isolation, i.e., respiratory versus body fluid isolation precautions.

In a multidisciplinary setting, all personnel must constantly monitor their own movements and practices, those of others, and the status of the overall field to prevent inadvertent breaks in sterile or clean technique. It is expected that personnel will alert other staff when the field or objects are potentially contaminated.

Health care workers can also promote asepsis by evaluating, creating, and periodically updating policies and procedures that relate to this principle.

Perioperative Theatre layout

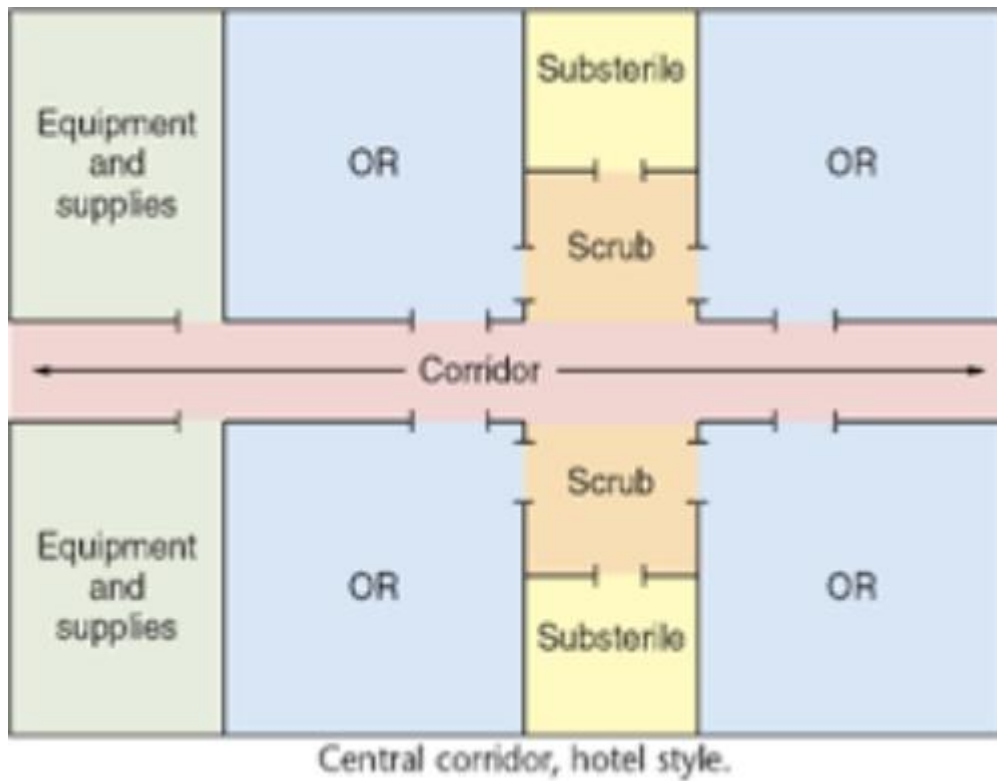
The adaptive layout for operating theatre (ALOT) problem in hospitals seeks to determine the ‘most efficient’ layout placement of a set of health-care operating facilities, corridors and elevators in a designated area subject to a set of constraints on professional standards.

Such standards include regulations on: hygiene, safety and security of stakeholders (doctors, medical staff, patients and visitors); movements of technologies; and specifications of operating rooms (functions, orientations, space sizes, and desired closeness). Existing ALOT layouts are mostly generated from designs based on experiential judgments of experts.

Due to the lack of scientific rigor and huge impact of layout design on the efficiency and effectiveness of an operating theatre, the paper proposes mixed integer linear programming models to find optimal layouts under three different design variants: ALOT with multiple sections; ALOT with multiple rows and ALOT with multiple floors.

Each variant has different demands for personnel, patients, and technologies over a planning horizon. Operating facilities can exchange functions at rearrangement costs from one period to another to meet the changing demands.

The general objective consists of two sub-objectives: the first sub-objective is to minimize the total sum of the rearrangement and travel costs whereas the second sub-objective is to maximize the total sum of desired closeness among facilities. Computational experiences are presented on a set of quasi-real data instances for a hospital in France. They demonstrate the effectiveness of the formulations in providing optimal layouts for realistic-sized instances. Conclusion and future research directions are presented.



Perioperative

Theatre instruments and apparatus

operating room equipment list for operation theatre Surgical Lamp Light. It has the wall mounted and Floor stand type, it shows the wall mounted type in the above picture. ...x-ray machine. ...Surgical table bed. ...patient monitor. ...Trolley. ...Anaesthesia machine. ...Infusion pump. ...Pendant.



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Perioperative Theatre supplies



Operating rooms (ORs) are specialized workspaces that require highly functional equipment of the highest quality, which surgeons and other OR personnel use to perform lifesaving procedures.

Without this equipment in place, OR professionals may struggle to deliver the care that patients need. There is a definite number of essential operating room equipment that OR professionals need on a per case basis. Each piece of equipment serves a distinct purpose and makes it easier for OR personnel to get the job done.

Here's a closer look at some of the most common required OR equipment that serve the needs of OR staff and patients everyday.

1. Utility Columns

Utility columns accommodate a variety of OR-administered services, ensuring better management of workspace in an OR. The columns can also be retractable or fixed and provide a viable solution for the distribution of communication units, electrical services and medical gases in OR suites.

With utility columns, surgeons can access electrical devices that are pre-wired, along with medical gas outlets that include a single point connection at the top of the unit.

The columns can even be integrated with a CO2 or nitrogen control panel to help surgeons deliver better patient support. Utility columns can be raised or lowered with ease.

When a utility column is raised, it provides additional space; when the column is lowered, it ensures that surgeons can hook up to various OR-administered services immediately.



2. Surgical and Exam Lights

surgical lights in the OR

Operating on a patient is challenging, and surgical lights empower medical professionals to work to the best of their ability.



Picking top-of-the-line surgical and exam lights ensure these professionals can enjoy the benefits of working in a well-lit OR at all times.

Many diverse surgical and exam lights currently are available, including:

- Ceiling-mounted surgical lights: Ceiling-mounted LED and halogen lights are among the top options available to brighten your OR.

- Mobile surgical lights: Is your OR in transition? In a temporary room, or for a short-term solution, you simply cannot go wrong with mobile lights. Mobile surgical lights are bright and powerful, providing the ideal combination for just about any OR.

- Surgical headlamps: Light, bright and comfortable, surgical headlamps provide surgeons with extra illumination right at the spot they are investigating. These headlamps are lightweight and comfortable and will illuminate just about any space immediately. Adding first-rate surgical and exam lights is necessary for an OR, regardless of a hospital's size. With these lights at your disposal, you'll be better equipped to maintain a secure, productive work environment.

3. Stretchers and Stretcher Accessories

Transporting a patient from one room to the next?

You'll be able to do so quickly and effortlessly if you deploy first-rate patient transfer stretchers for your OR. Hospital stretchers are crucial for various surgical procedures, such as ophthalmologic, bariatric and paediatric. They are designed to help patients remain comfortable and accessible. Stretcher accessories can customize a patient transport stretcher into a specific use stretcher. For example, Meditek manufactured a paediatric enclosure accessory for its line of QA3 stretchers, making the QA3 ideal for children with the combination of the enclosure keeping them safe and the stretcher's lowest height ability.

4. Cushions and Mattresses

Whenever a new patient arrives, you'll want to be ready to ensure that he or she can receive support in a plush or firm bed. Many cushions and mattresses are available that are designed to assist patients of all ages and sizes.

ideal mattresses in the OR

Mattresses and cushions can be purchased as complete sets or as separate leg, head, or body cushions. Two of the most common surgical table cushions include:

Standard Surgical Table Pads – As electrically conductive cushions, standard surgical table pads feature a supportive urethane core and Lectrolite covering that is simple to clean. The cushions also can help patients avoid bedsores and are ideal for patients during shorter medical procedures.

Premium Sof-Care Deluxe Surgical Table Pads – For superior pressure management and enhanced positioning capabilities, the Premium Sof-Care Deluxe surgical table pads represent great options. The surgical table pads feature an electrically conductive four-way anti-stretch cover for increased durability, fluid-resistant seams to reduce the risk of cross-contamination, and other features that make the table pads valuable in an OR.

Selecting the right cushions and mattresses – and ensuring they are available to patients every day – is crucial. With a dependable supplier of bedding at your side, you should have no trouble finding the ideal padding for your OR.

5. Space Management Booms

An everyday OR can become a stressful place to work if there is limited space. In fact, stress remains an ongoing problem among OR personnel, which is reflected in recent data. A study of OR personnel published by the *Thrita* medical journal indicated 48 percent of nurses said they experienced “work overload” that led to increased stress levels. Also, the study revealed OR technicians usually experienced higher stress levels than other OR personnel. Physical environment factors can affect a medical professional's stress levels. Comparatively, space management booms may help medical personnel conserve space, increase their productivity, and minimize their stress levels simultaneously. Space management booms can be added to almost any OR, regardless of size, and enable medical professionals to make the most of the available space in this room.

6. Sterilization and Cleaning Equipment

No one wants to put themselves or their patients at risk of infection, and sterilization and cleaning equipment is available that can make a world of difference for OR personnel and their patients. To better understand the importance of sterilization and cleaning equipment in ORs, let's consider the results of a recent study on OR foot traffic conducted by Orthopaedics.

The study, which examined “the number and length of door openings during nearly 200 knee and hip arthroplasty surgeries performed at Johns Hopkins Bayview Medical Centre over three months,” showed excessive door openings and foot traffic could put OR personnel and their patients at greater risk of infection. Study researchers found that OR doors were open during nearly one-third of knee and hip arthroplasty surgeries.

These open doors commonly defeated the purpose of many safety tactics that medical personnel used to keep ORs germ- and contamination-free, according to study researchers. “Our findings add to a growing body of evidence of a relatively common practice that could be a potential safety concern and raises questions about why doors get opened and how we can prevent or minimize the frequency and duration of behaviours that could compromise OR sterility,” study senior author Stephen Belkoff, Ph.D., said in a prepared statement.

“We know for sure is that there was a whole lot more traffic in and out of the OR than seems necessary or easily explained.” Sterilization and cleaning equipment, on the other hand, enables OR personnel to devote the necessary time and resources to maintain pristine work environments. This equipment allows medical professionals to protect themselves and their patients against a wide range of potential hazards, such as hospital acquired infections.

7. Disposables and Consumables

Facemasks and other disposables and consumables help safeguard an OR surgeon as well as team members and patients consistently. Many top-of-the-line disposables and consumables such as biohazard bags, patient care gloves, and waterproofing pads are available and designed for single use. That way, after you are finished using a single

disposable or consumable item, you'll be able to replace it for the next patient to improve your chances of maintaining a safe work environment.

8. Gel Pads and Positioners

Imagine what it would be like to provide patients with unparalleled support to help them combat general pain and soreness. Thanks to gel pads and positioners, you'll be able to do just that, as these items enable a patient to sit or rest comfortably and avoid skin or tissue damage. Gel pads and positioners typically offer a number of benefits for patients, including:

Can be heated or cooled instantly.

Repairable and reusable for long-lasting value.

Can't be affected by body fluids, and are easy to clean.

Serve as cost-effective options, making them great choices for hospitals of all sizes.

Leakproof, guaranteeing you won't have to worry about each product's interior spilling out.

9. Surgical Tables and Surgical Table Accessories

A high-quality surgical table and surgical table accessories are at the core of ensuring your OR team can perform the surgery at hand with confidence.

OR surgical tables high quality image

Surgical tables are valuable for surgeons during both open and minimally invasive procedures. The tables are manufactured for general surgery, neurology procedures, and other OR applications. Whether performing general or specialized surgeries, surgical table accessories will make either more efficient and comfortable for the patient.

A list of general and specific use accessories:

Anaesthesia Screens

Arm & Hand Surgery Tables

Arm Supports

Clamps

Drain Pans, Hoops & Screens

Head Supports

Leg Supports

Positioners

Restraint Straps

Side Extensions

Transfer Boards



Types of surgeries



There are various types of surgery and they can be classified based on different parameters

Based on Timing Surgery can be classified based on the urgency and need to perform the surgery.

Elective Surgery

The name “elective” might imply that this type of surgery is optional, but that’s not always the case. An elective procedure is simply one that is planned in advance, rather than one that’s done in an emergency situation. A wide range of surgical procedures can be considered elective surgery. Cosmetic surgeries fall into this category, but so can things like ear tubes, tonsillectomies, and scoliosis surgery. Although these procedures may be done “electively,” they can be significant and potentially life-changing operations.

Semi-elective Surgery

Semi-elective surgery is a surgery that must be done to preserve the patient’s life, but does not need to be performed immediately.

Urgent Surgery

An urgent surgery is one that can wait until the patient is medically stable, but should generally be done today or tomorrow

Emergency Surgery

An emergency surgery is one that must be performed without delay; the patient has no choice other than immediate surgery, if they do not want to risk permanent disability or death.

Based on Body Part

Surgery can also be classified based on the body part that is being operated on. When surgery is performed on one organ system or structure, it may be classed by the tissue, organ or organ system involved. Examples include:

Gastrointestinal surgery

This type of surgery is performed within the digestive tract and its accessory organs. Gastrointestinal surgery involves the treatment for diseases of the parts of the body involved in digestion. This includes the oesophagus, stomach, small intestine, large intestine, and rectum. It also includes the liver, gallbladder, and pancreas

Orthopaedic surgery

This is performed on bones or muscles. The medical specialty dedicated to the diagnosis, treatment, rehabilitation, and prevention of injuries of the musculoskeletal system including the muscles, bones, spine, joints, ligaments, tendons, and nerves.

Cardiovascular Surgery

Cardiovascular surgery is the surgical treatment of diseases and disorders of the heart and blood vessels. Examples of cardiovascular surgery include open heart surgery such as coronary artery bypass grafting, heart valve replacements, heart transplant and vascular surgery such as abdominal aortic aneurysm repairs and femoral-tibial bypass grafting.

Gynaecology

Gynaecology refers to the surgical specialty dealing with the health of the female reproductive system (uterus, vagina and ovaries).

Neurosurgery

The surgical specialty involved in the treatment of disorders of the brain, spinal cord, and peripheral nerves.

Ophthalmology (Eyes)

A medical specialty focusing on the diagnosis and treatment of disorders of the eye. Conditions include cataracts, glaucoma, and conjunctivitis, among others.

Oral and Maxillofacial Surgery

A dental specialty focusing on the diagnosis and surgical treatment of diseases, injuries and defects of the mouth, teeth and gums. Examples of oral surgery include removal of wisdom teeth, facial injury repair, cleft lip and palate repair, and repair of uneven jaws.

Otolaryngology – Ears, Nose and Throat (ENT)

Otolaryngology is the medical and surgical treatment of the head and neck, including ears, nose and throat.

Thoracic

This is also known as Cardiothoracic surgery. This is the field of medicine involved in surgical treatment of organs inside the thorax (the chest)—generally treatment of conditions of the heart (heart disease) and lungs (lung disease).

Urology

This is a type of surgery that is performed on the urinary tract. It is a surgical specialty that deals with problems affecting the urinary tract including kidneys, bladder, and prostate.

Based on Equipment Used

Surgery can also be classified based on the type of surgical equipment's used for the procedure.

Robotic Surgery

Robotic surgery makes use of a surgical robot, such as the Da Vinci or the Zeus surgical systems, to control the instrumentation under the direction of the surgeon. Da Vinci's advanced level of technology takes surgery beyond the limits of the human hand. This da Vinci Surgical System compliments our goal of extending minimally invasive surgery to the broadest base of patients including but not limited to: cholecystectomy (gallbladder removal), hysterectomies, myomectomies, hiatal and hernia repairs, removal of one or both adrenal glands, GERD related diagnoses (Reflux), and oesophageal stricture.

The da Vinci Surgical System offers

Enhanced 3D, high-definition vision of operative field, with up to 10x magnification

New optimal dual console, which allows a second surgeon to provide assistance

Ability to mimic the surgeon's hand movements with enhanced dexterity

Microsurgery

Microsurgery involves the use of an operating microscope for the surgeon to see small structures.

The most obvious developments have been procedures developed to allow anastomosis of successively smaller blood vessels and nerves (typically 1 mm in diameter) which have allowed transfer of tissue from one part of the body to another and re-attachment of

severed parts. Microsurgical techniques are utilized by several specialties today, such as Replantation, Transplantation, treatment of infertility, etc

Laser Surgery

Laser surgery involves use of a laser for cutting tissue instead of a scalpel or similar surgical instruments.

Laser surgery is commonly used on the eye. Techniques used include LASIK, which is used to correct near and far-sightedness in vision, and photorefractive keratectomy, a procedure which permanently reshapes the cornea using an excimer laser to remove a small amount of the human tissue.

Based on degree of invasiveness

Surgery can also be classified based on the degree of invasiveness of the surgical procedure. By invasion, this refers to the medical procedure in which the part of the body is entered by incision.

Minimally Invasive Surgery

Minimally invasive surgery is any technique involved in surgery that does not require a large incision. This relatively new approach allows the patient to recuperate faster with less pain. Not all conditions are suitable for minimally invasive surgery. Many surgery techniques now fall under minimally invasive surgery: Laparoscopy, Endoscopy, Arthroscopy and many others.

Open Surgery

An “open” surgery means the cutting of skin and tissues so that the surgeon has a full view of the structures or organs involved. Examples of open surgery are the removal of the organs, such as the gallbladder or kidneys.

Based on purpose

Surgery can also be classified based on the purpose for which the surgery is being performed. Surgeries are performed for various purposes and such can be classified based on the purpose for the surgery.

Exploratory surgery

Exploratory surgery is a diagnostic method used by doctors when trying to find a diagnosis for an ailment. It is used most commonly to diagnose or locate cancer in humans.

Therapeutic surgery

Therapeutic surgery treats a previously diagnosed condition

Cosmetic surgery

Cosmetic surgery is where a person chooses to have an operation, or invasive medical procedure, to change their physical appearance for cosmetic rather than medical reasons.

In conclusion, there are different types of surgery and they are classified based on a number parameters ranging from timing, body parts, equipment and purpose of the surgery.

Decontamination

Decontamination is the process of removing or neutralizing chemical or biological agents so that they no longer pose a hazard. For military purposes, decontamination is undertaken to restore the combat effectiveness of equipment and personnel as rapidly as possible. Most current decontamination systems are labour and resource intensive, require excessive amounts of water, are corrosive and/or toxic, and are not considered environmentally safe. Current R&D is focused on developing decontamination systems that would overcome these limitations and effectively decontaminate a broad spectrum of CB agents from all surfaces and materials.

R&D on decontamination has evolved over the years. Ideally, these efforts would result in the identification and deployment of a "universal" decontaminating system. However, this has not happened. During and after the 1950s, most decontamination involved aqueous-based systems or the nonaqueous Decontamination Solution Number 2 (DS2), both of which have weaknesses. Aqueous-based systems require large amounts of water (often not available on a battlefield) and are very inefficient at removing agent that has been adsorbed onto or absorbed into a painted surface. In addition, the contaminated water must be gotten rid of. DS2 is too corrosive for many applications. Ongoing attempts to develop dry systems

(referred to euphemistically as "magic pixie dusts") that would be easy to transport and use, would not require large amounts of fluids, and could replace DS2 or aqueous systems, have not yet been successful.

Fixed sites that are critical to the deployment of troops have now become attractive and vulnerable targets for CB attack, increasing the importance of protecting them or, if protection is impossible, improving our ability to restore contaminated sites and equipment to operational status. The focus of R&D has, therefore, been shifted from the decontamination of mobile forces to the decontamination of fixed sites. The key differences between decontamination of fixed sites and mobile forces are summarized

Basic anaesthesia

Basic anaesthetic techniques

General anaesthesia (GA)

As a novice trainee, general anaesthesia is the most common type of anaesthetic that you will be required to provide and become competent with as part of the IAC.

General anaesthesia consists of a triad of:

unconsciousness

analgesia

+/- muscle relaxation.

A general anaesthetic (GA) can be subdivided into 3 key stages:

induction - getting the patient to sleep

maintenance - keeping the patient asleep during the operation

emergence - waking the patient up at the end of the procedure.

The most common techniques you will be exposed to are:

spontaneous ventilation with a laryngeal mask airway (GA LMA SV)

intermittent positive pressure ventilation with an endotracheal tube (GA ETT IPPV)

Rapid Sequence Induction to protect the airway if there is a risk of soiling from gastric contents (RSI GA ETT IPPV).

Your choice of technique will be influenced by specific patient, surgical and anaesthetic factors, as well as the level of your experience in the specialty and its related subspecialties.

The sections that follow will give you a broad overview and the basic principles underpinning these three techniques.

General anaesthesia: spontaneous ventilation with an LMA

Generally used for elective or emergency surgery in patients who have been fasted and where there is no risk of aspiration of gastric contents. Common risk factors for aspiration include:

history of reflux or hiatus hernia intra-abdominal pathology pregnancy recent major trauma or administration of opiates morbid obesity autonomic dysfunction associated with diabetes.

Suitable for most peripheral surgery in patients who are not obese and for some minor abdominal surgery where muscle relaxation is not needed (i.e. elective inguinal hernia repair).

Can be used for patients in the supine or lateral position. Not recommended for use in very long cases. Experienced anaesthetists may use this technique in ENT and Maxillofacial surgery for short procedures. Basic principles in ASA 1/2 patient: attach basic monitoring (SpO₂, ECG and NIBP) pre-oxygenate the patient

induction - intravenous (propofol) or inhalational induction (sevoflurane) + small dose of fentanyl followed by insertion of LMA

maintenance - oxygen/air or oxygen/nitrous oxide plus volatile or intravenous agent

emergence - 100% Oxygen and turn off volatile or intravenous agent. Remove LMA when patient wakes up.

General anaesthesia: endotracheal intubation and positive pressure ventilation

Generally used for elective or emergency surgery in patients who have been fasted and where there is no risk of aspiration of gastric contents but where:

muscle relaxation is required to facilitate surgery

surgery will increase intra-abdominal pressure or the bowel will be manipulated

Used for elective or emergency surgery in patients who are either not fasted or where there is a risk of aspiration of gastric contents. The airway is protected from aspiration by the application of cricoid pressure prior to rapid intubation using a depolarizing muscle relaxant. Thereafter, the anaesthetic technique is the same as an ETT with IPPV in the previous section.

Basic principles in ASA 1/2 patient: attach basic monitoring (SpO₂, ECG and NIBP) pre-oxygenate the patient for 3 minutes proper preparation – tipping trolley, suction available, equipment checked induction intravenous (thiopental or propofol) suxamethonium to facilitate intubation with cricoid pressure applied by anaesthetic assistant to prevent aspiration oral intubation and confirmation of tube placement by visualization, auscultation and capnography release of cricoid pressure non-depolarizing muscle relaxant to maintain muscle relaxation maintenance oxygen/air or oxygen/nitrous oxide plus volatile agent top up doses of muscle relaxant guided by neuromuscular monitoring additional analgesia as required emergence 100% oxygen turn off volatile or intravenous agent reverse neuromuscular block remove ETT when patient spontaneously breathing, regains airway reflexes and wakes up it is recommended that this is performed with the patient in the left lateral position. Other types of anaesthesia Local infiltration Good choice for skin and soft-tissue surgery of limited size, depth and duration, e.g. ring block for a distal finger operation.

Regional anaesthesia Achieved by introducing a local anaesthetic agent near the nerves supplying that region. May be used as the sole mode of anaesthesia or combined with general anaesthesia or sedation

Basic pharmacology in theatre

DRUGS ACTING ON THE CENTRAL NERVOUS SYSTEM

To facilitate the understanding of the pharmacological and unwanted effects of CNS drugs, the physiological functions of the main CNS neurotransmitters are discussed briefly. Noradrenaline. Noradrenergic transmission is important in control of mood (functional deficiency resulting depression) controlling wakefulness, and alertness. Dopamine. Dopamine is important in motor control (Parkinsonism is due to dopamine deficiency), has behavioural effects (excessive dopamine activity is implicated in schizophrenia), important in hormone release (prolactin, GH) and dopamine in chemoreceptor trigger zone causes nausea and vomiting. 5-HT.

Physiological functions associated with 5-HT pathways include; feeding behaviour, behavioural response (hallucinatory behaviour), control of mood and emotion, control of body temperature and vomiting. Acetylcholine (ACh).

ACh has effects on arousal, on learning, and on short-term memory. Dementia and parkinsonism are associated with abnormalities in cholinergic pathways. GABA. GABA is an inhibitory neurotransmitter in CNS. Glycine. is an inhibitory neurotransmitter, acts on GABA like receptor in the spinal cord?

GENERAL ANESTHETICS

General anesthesia involves the physiological changes: Reversible loss of response to painful stimuli, loss of consciousness and loss of motor and autonomic reflexes. Loss of consciousness is associated with inhibition of the activity of reticular formation. General anesthetics are administered by inhalation or by intravenous routes. They are classified into two on the basis of their route of administration as inhalation and intravenous anesthetics.

Inhalation anesthetics.

The main agents are: Halothane, nitrous oxide, enflurane and ether.

1. Halothane: Is the most widely used agent, highly lipid soluble, potent. It causes arrhythmia,

hangover and the risk of liver damage is high if used repeatedly.

2. Nitrous oxide: Odourless and colourless gas. It is rapid in action and also an effective analgesic agent. Its potency is low, hence must be combined with other agents. It is a relatively free of serious unwanted effects.

3. Enflurane: Halogenated ether (similar to halothane). Poorly metabolized in the liver, thus less toxic than halothane. It is faster in its action, less liable to accumulate in the body fat compared to halothane. It causes seizure during induction and following recovery from anaesthesia.

4. Ether: Has analgesic and muscle relaxant properties. It is highly explosive, causes respiratory tract irritation, postoperative nausea and vomiting. It is not widely used currently.

INTRAVENOUS ANESTHETICS

Intravenous anesthetics act much more rapidly, producing unconsciousness in about 20 seconds, as soon as the drug reaches the brain from the site of its injection. These agents used for induction of anaesthesia followed by inhalation agent. The main induction agent in current use is: thiopentone, etomidate, propofol, ketamine and short acting benzodiazepine (midazolam).

Thiopentone: Thiopentone is a barbiturate with very high lipid solubility. After intravenous administration the drug enters to tissues with a large blood flow (liver, kidneys, brain, etc) and

more slowly to muscle. Uptake into body fat occurs slowly because of the low blood flow to this

tissue, which may cause prolonged effect if given repeatedly. It causes cardiovascular

depression. Etomidate: It is more quickly metabolized and the risk of cardiovascular depression is less compared to thiopentone. Etomidate suppresses the adrenal cortex, which has been associated with an increase in mortality in severely ill patients.

Ketamine: acts more slowly than thiopentone and produces a different effect, known as

dissociative anaesthesia in which there is a marked sensory loss and analgesia, as well as

amnesia and paralysis of movement, without actual loss of consciousness. Ketamine causes

dysphoria, hallucinations during recovery. Benzodiazepines including diazepam, lorazepam, and midazolam are used in general anesthetic procedures. Compared with intravenous barbiturates, benzodiazepines produce a slower onset of central nervous system effects. Benzodiazepines prolong the postanesthetic recovery period but also cause a high incidence of amnesia for events occurring after the drug is administered. The benzodiazepines are useful in anesthesia as premedication and intraoperative sedation.

Opioid analgesic anesthesia: Opioid analgesics can be used for general anesthesia, in patients undergoing cardiac surgery and fentanyl and its derivatives are commonly used for these purposes.

Preanesthetic medication: It is the use of drugs prior to the administration of anaesthetic agent with the important objective of making anaesthesia safer and more agreeable to the patient. The drugs commonly used are, opioid analgesics, barbiturates, anticholinergics, anti emetics and glucocorticoids.

SEDATIVE AND HYPNOTIC DRUGS

Anxiolytic drugs are used to treat the symptoms of anxiety, whereas hypnotic drugs are used to treat insomnia. The same drugs are used for both purposes. Classes of anxiolytic and hypnotic drugs: The main groups of the drugs are: 1. Benzodiazepines. Benzodiazepines are the most important group, used as sedative and hypnotic agents. 2. 5-HT_{1A} receptor agonist (e.g. buspirone). It is recently introduced anxiolytic. 3. Barbiturates (phenobarbitone).

They are nowadays less commonly used as sedativehypnotics.4. β -adrenoceptor antagonists (e.g. propranolol). They are used to treat some forms of anxiety,where physical symptoms (sweating, tremor, and tachycardia), are troublesome. They are not used as hypnotics. 5. Miscellaneous drugs (chloral hydrate, paraldehyde, and diphenhydramine). These drugs are not commonly recommended for axiety or insomnia.Benzodiazepines Benzodiazepines are well absorbed when given orally. They bind strongly to plasma proteins,however, many of them accumulate gradually in the body fat (i.e. they are highly lipid soluble).Benzodiazepines are inactivated by the liver and excreted in the urine.Based on their duration of action roughly divided into short acting (flurazepam, triazolam),medium acting (alprazepam, lorazepam) and long acting compounds (diazepam,chlordiazepoxide, clonazepam).Pharmacodynamics Act by binding to a specific regulatory site on the GABAA receptor, thus enhancing the inhibitory effects of GABA. Central nervous system effects of benzodiazepines include:

1. Reduction of anxiety and aggression.
2. Sedation and induction of sleep.
3. Reduction of muscle tone and coordination.
4. Anticonvulsant effects.

Clinical Uses

- Treatment insomnia
- Anxiety
- Preoperative mediations
- Acute alcohol withdrawal
- As anticonvulsants
- Chronic muscle spasm and spasticity

Unwanted effects

- Toxic effects due to acute overdosage causes prolonged sleep.

- Unwanted effects occurring during normal therapeutic use includes: drowsiness,

confusion, amnesia, and impaired motor coordination. Tolerance and dependence: Pharmacokinetic and tissue tolerance and also cause physical dependence. i.e. stopping benzodiazepines treatment after weeks or months causes an increase in symptoms of anxiety.

5-HT_{1A} receptor agonist Buspirone is a potent agonist of 5-HT_{1A} receptors. Anxiolytic effects take days to weeks to develop. Buspirone does not cause sedation, motor incoordination and withdrawal effects. The main side effects are nausea, dizziness, headache, and restlessness.

Barbiturates

They are non-selective CNS depressants, which produce effects ranging from sedation and reduction of anxiety, to unconsciousness and death from respiratory and cardiovascular failure.

Barbiturates act by enhancing action of GABA, but less specific than benzodiazepines. They are potent inducers of hepatic drug metabolizing enzymes, hence likely to cause drug interaction.

Tolerance and dependence occur, more than benzodiazepines.

ANTIEPILEPTIC DRUGS

Seizure is associated with the episodic high frequency discharge of impulses by a group of neurons in the brain.

Seizure may be partial or generalized depending on the location and the spread of the abnormal neuronal discharge. The attack mainly involves motor, sensory or behavioural phenomena.

Partial seizures are often associated with damage to the brain, whereas generalized seizure occurs without obvious cause. Two common forms of generalized seizures are grand mal and

petit mal.

Mechanism of action

Anticonvulsant drugs act by two mechanisms: by reducing electrical excitability of cell membrane and by enhancing GABA mediated synaptic transmission.

The main drugs used in the treatment of epilepsy are phenytoin, carbamazepine, valproate, ethosuximide and phenobarbitone.

Phenytoin

It is commonly used antiepileptic drug. It is effective against different forms of partial and generalized seizures; however it is not effective in absence seizures.

Well absorbed when given orally. It is metabolised by the liver. It is liver enzyme inducer and

therefore, increases the rate of metabolism of other drugs. Main side effects are sedation, confusion, gum hyperplasia, skin rash, anaemia, nystagmus, and diplopia.

Carbamazepine

It is derived from tricyclic antidepressant. Its pharmacological action resembles those of phenytoin, however, it is chiefly effective in the treatment of partial seizure. It is also used in the

treatment of trigeminal neuralgia and manic-depressive illness.

It is powerful inducer of liver microsomal enzymes, thus accelerates the metabolism of phenytoin, warfarin, oral contraceptives and corticosteroids.

Carbamazepine causes sedation, mental disturbances and water retention. Valproate

Valproate is chemically unrelated to the other antiepileptic drugs. The mechanism of action is

unknown. It is used in grand mal, partial, petit mal and myoclonic seizure. Relatively has few side effects, however, it is potentially hepatotoxic. It is non sedating.

Ethosuximide

Has fewer side effects and used in the treatment of absence seizures.

Phenobarbitone

It is well absorbed after oral administration and widely distributed. Renal excretion is enhanced

by acidification of the urine. Phenobarbitone is liver enzyme inducer and hence accelerates the

metabolism of many drugs like oral contraceptives and warfarin.

The clinical use of phenobarbitone is nearly the same as that of phenytoin. The most important

unwanted effect is sedation.

Benzodiazepines: Clonazepam and related compounds, clobazam are claimed to be relatively

selective as antiepileptic drugs. Sedation is the main side effect of these compounds, and an

added problem may be the withdrawal syndrome, which results in an exacerbation of seizures if

the drug is stopped.

Perioperative Theatre ethics

Medico–legal aspect of perioperative theatre care

Introduction

A basic principle of patient care is to do no further harm. A health care provider usually avoids legal exposure if he or she acts:

According to an appropriate standard of care

Ethics and law

Law - a body of principles recognized and applied by the state in the administration

Ethics - Ethics are quite distinct from legal law, although these do overlap in important ways

Laws applicable to nursing

Public Health Act cap 242 implies that the people of Kenya are entitled to properly secured and maintained health care services which are available resources of the state

Right to considerate and respectful care from all members of the health care system at all times and under all circumstances Laws applicable to patient care

Patients of bills of right to information (disclosure): preventive health care including health education, Individual own diagnosis and specific treatment program

Right to informed choices, participation in treatment opinion and decisions Right to access emergency services when and where the need arises. Patients of bill of rights Right to privacy, dignity and comfort Right to

Right to

Confidentiality safety

Right to expect reasonable coordination and continuity of care i.e. appointment dates, and Drs. who will be continuing with care after is discharge Medical negligence and malpractice

Negligence Unintentional breach of duty or failure Malpractice Held liable for negligence Essential elements

Duty and standard of duty, causation injury care, breach of & damage

Negligence

Deviation from accepted or expected standards of care expected to protect from unreasonable risk of harm or the failure to use the care/skills that any care giver in the same/similar situation would be expected to do

Occurs when there is duty care

Carelessness assumes the legal quality of negligence

Malpractice

Professional negligence when the conduct fails to meet the legal standard of care and someone is damaged as a result Law has developed a measuring scale called standard of care

Malpractice

Standard of care – the expected care, skill, & judgment under similar circumstances by a similarly trained, reasonable nurse or what a reasonably prudent person acting under similar would do

Elements of negligence

Failure to meet the standards of due care Any harm resulting from failure to meet the standard Breach of this standard causes an injury to the patient Abandonment of patients under your care •Failure to observe & take appropriate action Leaving patients unattended /refusing to attend to a patient Elements of negligence

Injury sustained to a patient under your care Damage -Derived from harm or injury Nosocomial infection- meaning that there is no proper measure of infection control

Loss of patients property/documents •Documentation: Omission & Commission

Wrongful death - patient dies because of negligence of health care organization or health care professionals mistakes

Elements required to prove negligence Duty of Care (Duty to Act)

Duty of care to the patient (Nurses assume a duty to provide care for the patient which is consistent with the standard of Care)

Knowledge of the case & management so as to provide reasonable care - need to collaborate with other professionals

Consideration of professional ethics Elements required to prove negligence Breach of Duty

Failure to perform to the standard or failure to act consistently with applicable standard of care leading to injury and damages Damages

Actual loss which occurred or harm Medical-legal cases

Is any case where the discipline of medicine comes to help the legal fraternity in its discharge of duties

Interface of dimensional perplexing medicine & can be & law is multi quite complex &

H/Care professionals need to be very cautious in dealing with the medico- legal cases

Medical-legal cases

Request of the patient, relatives or friend for not registering the case as medico-legal should not be accepted

Medico-legal case should be registered as soon as the Dr. suspects foul play or case brought several days after the incident

Example of medical legal cases

All cases of injuries

All cases of burns

Alleged cases of assault

All cases of suspected or evident of poisoning or intoxication

Cases of unconscious/comatose where its cause is not natural or not clear

Cases of suspected self-inflicted injuries or attempted suicide

End of life issues

Life sustaining measures

Cardio-pulmonary resuscitation (CPR) ~ When to stop mechanical ventilation uEnd-of-life decisions

With-holding or Withdrawing Treatment

Do not resuscitate (DNR) – Euthanasia

End of life issues

Pt must be made aware about their rights to: u Initiate directives & encouraged to do so

Consent or refuse Rx

Discussions regarding advance directives/end - of-life issues should be:-

made as early as possible before death is imminent

Not be made in hospital setting to allow more time for discussion, processing & decision making

Advance directives

Communication that specifies a person's preference about medical care in the event that person becomes incapacitated

Includes

Do not resuscitate(DNR)order

The pt. or surrogate has given clear directives not to resuscitate the pt. in event of cardiopulmonary arrest

Advance directives

There should be discussions, broad consultation and accurate documentation that reflect discussions between family and members of the critical care team and any subsequent decisions.

Any directives those involved

should be clear to all in patients care

Advance directives

Do not resuscitate(DNR)order cont'd

The order is clearly written in the pt.'s records by physicians so that misunderstanding do not occur It must meet the facility's policy Allow-a-natural death In what situations should CPR be used? Done to patients with irreversible or terminal illness

Advance directives

Do not resuscitate(DNR)order cont'd uHow long should efforts continue?

A generally accepted Position is that resuscitation should cease if the physician determines the efforts to be futile or hopeless

Advance directives

Living will

Provide mechanisms in which an individual can allow which specific Rx to be withheld in an event they become incapacitated

May not always be legally binding uCan be changed from time to time

Advance directives

Health care proxy or surrogate/durable power of attorney for health care

Pt legally designate a person whom they trust to make decisions on their behalf should they become incapacitated

Protects pt.'s interest medically than a living will

End of life decisions

End-of-life decisions are often difficult & are best made after careful discussions between the H/Care professional & the patient (or the legal next of kin)

End of life decisions

Withholding or withdrawal of treatment

This involves the withholding or withdrawal of multiple life sustaining therapies

Withholding" refers to not initiating treatment, whereas "withdrawing "refers to stopping treatment once started

End of life decisions

Effective Management of Surgical Instruments Withdrawing of life support is ethically acceptable and clinically desirable if it reduces the unnecessary suffering in patients whose life is considered hopeless and any treatment given is futile e.g. withdrawing of ventilation support, ionotropic drugs etc.

Withdrawing Tx for sound moral reasons does not violate professional obligations; however H/Care professionals may find it emotionally difficult to withdraw Tx

End of life decisions

The nurse ensures that the patient/legal next of kin understands the information by clarifying technical terms & helping the patient/next of kin weigh treatment options

End of life decisions

Euthanasia t's illegal Has primary intention of intervention e.g. giving a lethal injection to cause death,Can be tested in court

Assignment 2

Discuss the differences between the following terms

1. Tort

2. Libel vs Assault vs slander

3. Breach of contract vs absconding duty 4. Negligence vs malpractice

EMPLOYMENT ACT CHAPTER 226

EMPLOYMENT RELATIONSHIP

Contracts Contract of service

No person shall be employed under a contract of service except in accordance with the provisions of this Act. Oral and written contracts

The provisions of this Act shall apply to oral and written contracts.

General provision of contract of service A contract of service—

(a) for a period or a number of working days which amount in the

aggregate to the equivalent, of three months or more; or

(b) which provides for the performance of any specified work which could not reasonably be expected to be completed within a period or a number of working days amounting in the aggregate to the equivalent of three months, shall be in writing.

Employment particulars

(1) A written contract of service specified in section 9 shall state particulars

of employment which may, subject to subsection (3), be given in instalments and

shall be given not later than two months after the beginning of the employment.

(2) A written contract of service shall state—

(a) the name, age, permanent address and sex of the employee; (b) the name of the employer;

(c) the job description of the employment;

(d) the date of commencement of the employment; (e) the form and duration of the contract;

(f) the place of work; (g) the hours of work;

(h) the remuneration, scale or rate of remuneration, the method of calculating that remuneration and details of any other benefits;

(i) the intervals at which remuneration is paid; and

(j) the date on which the employee's period of continuous employment began, taking into account any employment with a previous employer which counts towards that period; and

(k) any other prescribed matter.

Payment, disposal and recovery of wages, allowances,

(1) Subject to this Act, an employer shall pay the entire amount of the wages

earned by or payable to an employee in respect of work done by the employee in

pursuance of a contract of service directly, in the currency of Kenya— (a) in cash (b) into an account at a bank, or building society, designated by the employee;

(c) by cheque, postal order or money order in favor of the employee; or

Payment etc..

(1) In the absence of an employee, to a person other than the employee,

if the person is duly authorized by the employee in writing to receive the wages on the employee's behalf. (2) An employer shall pay wages to an employee on a working day, and during

working hours, at or near to the place of employment or at such other place as may

be agreed between the employer and the employee.

(3) An employer shall not pay wages to an employee in any place where intoxicating liquor is sold or readily available for supply, except in the case of employees employed to work in that place.

(4) No person shall give or promise to any person any advance of money or any valuable consideration upon a condition expressed or implied that the person or any dependent of that person shall enter upon any employment.

Sexual harassment

An employee is sexually harassed if the employer or that employee or a representative of that employer or a co-worker—

(a) directly or indirectly requests that employee for sexual intercourse, sexual contact or any other form of sexual activity that contains an implied or express—

promise of preferential treatment in employment; threat of detrimental treatment in employment; or threat about the present or future employment status of the employee;

(b) uses language whether written or spoken of a sexual nature; (c) uses visual material of a sexual nature; or

(d) shows physical behavior of a sexual nature which directly or indirectly subjects the employee to behavior that is unwelcome or offensive to that employee and that by its nature has a detrimental effect on that employee's employment, job performance, or job satisfaction.

Sexual harassment policy i. that every employee is entitled to employment that is free of

ii. sexual harassment;

iii. that the employer shall take steps to ensure that no employee is subjected to sexual harassment;

iv. that the employer shall take such disciplinary measures as the employer deems appropriate against any person under the employer's direction, who subjects any employee to sexual harassment;

v. explaining how complaints of sexual harassment may be brought to the attention of the employer; and that the employer will not disclose the name of a complainant or the circumstances related to the complaint to any person except where disclosure is necessary for the purpose of

vi. investigating the complaint or taking disciplinary measures in relation thereto.

Leave 1. Annual Leave

According to section 28 of the Kenyan Employment Act, every employee is entitled to Annual leave days after every 12 consecutive months of service with full pay. As an employer, you can divide these 21 leave days in intervals with the consent of your employee.

2. Sick Leave

Section 30 of the employment ACT highlights that an employee who has worked for you for 2 consecutive months is entitled to sick leave of not less than seven days with full pay.

3. The maternity /Paternity leave

Further Section 29 of the ACT says that a female employee is entitled to 3 months maternity leave with full pay after which she also has a right to return to her job or with better terms than what she had before she left for the maternity leave. Clause 8 of section 29 provides that male employees are as well entitled to two weeks paternity leave with full pay.

4. Compassionate leave

Compassionate leave in Kenya allows an employee to attend to his/her misfortunes which are usually not planned for and may occur at any point in time including death, sickness or accidents of relatives and friends.

5. Compulsory Leave

Compulsory leave is an administrative leave imposed on an employee by the employer normally to pave way for investigations into possible employment offences which may ultimately lead to commencement of disciplinary proceedings.

6. Leave of absence

Upon request, other employers in Kenya allow their employees to unpaid leave of absence from work for a period of time depending on the organization.

Identification and reporting of faulty Operation Theatre apparatus

MANAGING SURGICAL INSTRUMENTS IS A TEAM EFFORT

Effective Management of Surgical Instruments

Perioperative personnel and processing personnel must WORK TOGETHER

to ensure the effective management of surgical instruments. Effective Management of Surgical Instruments

PURPOSE STATEMENT

The purpose of this educational activity is to demonstrate proper management of surgical instruments and to emphasize that effective communication between perioperative and instrument processing personnel is essential to ensure that instruments used in surgical and other invasive procedures facilitate safe and effective patient outcomes.

Effective Management of Surgical Instruments

PATIENT SAFETY

- Optimal surgical outcome.
- Reduced complications related to tissue damage.
- Reduced incidence of infection.
- Reduced recuperation time

Effective Management of Surgical Instruments

REDUCED EXPENSES

- Reduce patient length of stay.
- Reduce expenses related to hospital-associated infection.
- Reduce expenses related to complications.
- Extend instrument life.

- Reduce replacement costs.
- A major cause of instrument damage is mishandling and abuse.

Effective Management of Surgical Instruments

Instrument management

Begins in the Operating ROOM at the point of instruments

DURING THE PROCEDURE

Instruments must be

INSPECTED BEFORE use on the field.

Effective Management of Surgical Instruments

Any DEFECTIVE OR MISSING INSTRUMENTS must be CLEARLY IDENTIFIED for SPD or the instrument technician in ambulatory surgery. Effective Management of Surgical Instruments

HANDLING

- Handle instruments carefully and ONE AT A TIME or in SMALL NUMBERS.
- Use instruments ONLY for the purpose for which they were intended

Effective Management of Surgical Instruments

- Use scissors only for the MATERIAL FOR WHICH THEY WERE DESIGNED.

Cutting other materials can cause misalignment, loss of sharpness, damage (e.g. Use Mayo scissors, not Metzenbaums or more delicate scissors to cut drapes and material other than tissue).

- Use sharp instruments (diamond knives, cystotomes, dissectors, knives) ONLY FOR THEIR INTENDED USE.

Misuse can cause blades and edges to dull and impair surgical performance.

Effective Management of Surgical Instruments

- ~ Use forceps APPROPRIATE TO THE TASK at hand

Improper alignment can create serious problems with function and harm to the patient

- MATCH NEEDLE HOLDERS TO THE SIZE OF NEEDLE for which it is intended

–Large needles will spring the jaws of delicate needle holders and reduce holding power

Effective Management of Surgical Instruments

- During the procedure, wipe blood and tissue from instruments

IMMEDIATELY AFTER USE.

Debris that is allowed to dry on instruments causes deterioration, corrosion, and pitting.

Saline solution may be used to wipe instruments that will be used again during the procedure.

Effective Management of Surgical Instruments

- PROTECT TIPS of instruments with tip protectors.

Prevent tips of instruments from contacting other instruments and snagging on towels and sponges

Effective Management of Surgical Instruments

FOLLOWING THE PROCEDURE

- Remove disposable sharps; manage sharps appropriately.

Effective Management of Surgical Instrument

PREPARE THE INSTRUMENTS

- Open box locks.

Insure that all multipart instruments are DISASSEMBLED.

KEEP PARTS TOGETHER for easy reassembly after processing

EXPOSE all surfaces

Cleaning, disinfectants, and sterilants must

CONTACT ALL SURFACES of an instrument for the process to be effective

Effective Management of Surgical Instruments

- All cannulated equipment and handpieces should be FLUSHED

Effective Management of Surgical Instruments

Organize instruments into sets.

- All instruments, used and unused, go back into the set.
- Once opened onto the sterile field, the entire set is considered contaminated.
- Place HEAVY INSTRUMENTS ON THE BOTTOM; lighter instruments on top.

Effective Management of Surgical Instruments

- Clearly identify deficient/missing instruments.

Effective Management of Surgical Instruments

- CONTAIN CONTAMINATED instruments to protect personnel.
- DO NOT ALLOW BLOOD AND DEBRIS TO DRY on instruments.
- Soak in demineralized water to prevent drying of bioburden.
- Saline, disinfectants, and chlorinated solutions can cause pitting and corrosion and should never be used for soaking instruments.
- Transport instruments to SPD as quickly as possible.

- Instruments should not remain in water for lengthy periods of time.
- Biofilms may form, particularly within lumens.

Effective Management of Surgical Instruments

BIOFILM

- An aggregate of organisms encased within a gelatinous matrix.
- Biofilm is the SLIMY FEELING on the inside of a container that has held water for a period of time.
- Once formed, a biofilm CANNOT BE RINSED AWAY; it can only be removed by mechanical means.
- The presence of a biofilm makes cleaning more difficult.
- Biofilm can pose a SERIOUS THREAT TO HEALTH.
- Biofilm CAN COMPROMISE THE DISINFECTION AND STERILIZATION PROCESSES

A biofilm CAN BREAK FREE from the surface of a device resulting in a MASSIVE INFUSION OF BACTERIA that can cause an infection that is difficult to treat.

Effective Management of Surgical Instruments

- Cleaning is the FIRST AND MOST IMPORTANT STEP in instrument processing.
- An instrument must be clean before it can be sterilized.
- Debris prevents sterilant from contacting all surfaces of an instrument.
- Improper cleaning of instruments can cause harm to patients.

AN INSTRUMENT CAN BE CLEAN BUT NOT STERILE; NO INSTRUMENT CAN BE STERILE BUT NOT CLEAN.

Effective Management of Surgical Instruments

STERILE PROCESSING RESPONSIBILITIES

- Cleaning/ Decontamination
 - Wash
 - Rinse
- Lubrication
- Inspection and Testing
 - Continue Reprocessing
 - Send for Repair/Discard
- Prepare for sterilization
- Sterilization
- Storage

Effective Management of Surgical Instruments

TOXIC ANTERIOR SEGMENT SYNDROME (TASS)

- An ACUTE INFLAMMATION of the anterior chamber, or segment, of the eye following cataract surgery.
- Can be caused by INADEQUATE OR INAPPROPRIATE CLEANING of instruments leaving irritants on the surface of intraocular surgical instruments.

Effective Management of Surgical Instrument

TOXIC ANTERIOR SEGMENT SYNDROME (TASS) (CONT.)

- Risks may be associated with the use of enzymatic cleaners.

- INCOMPLETE RINSING, especially of cannulated instruments, has been identified with TASS.
- Chemical sterilization methods such as EtO and peracetic acid may leave CHEMICAL RESIDUES on the instruments that promote TASS

Effective Management of Surgical Instruments

MANUAL CLEANING

- HAND WASH microsurgical instruments and trays.
- Remove mats from trays and CLEAN EACH SEPARATELY.
- Do not put delicate microsurgical instruments in mechanical washer unless it has a DELICATE CYCLE.

Effective Management of Surgical Instrument

- OPEN ALL Hinged microsurgical instruments for maximum exposure of hinges.
- Debris left in box locks or crevices can be baked on in sterilization, causing future Breakage UNDER STRESS.
- DISASSEMBLE instruments with removable parts Effective Management of Surgical Instruments
- Wash each instrument individually to ensure proper cleaning and to prevent injury.

Effective Management of Surgical Instruments

- Do not use abrasive cleaners when cleaning microsurgical instruments; they can damage or scar finishes.
- Use soft brush and instruments wipes.
- Brush all serrations, crevices, tips, handles, and hinges.
- Brushing should be done under the surface of the water to prevent aerosolization of contaminants.

Effective Management of Surgical Instruments

CANNULATED INSTRUMENTS

- Cannulas, irrigation cannulas, irrigation/aspiration (I/A) handpieces, suction tips, cystotomes.
- Use a brush of the appropriate size to clean the lumen; hold instrument and brush below the water surface.
- Clean and sterilize the brush DAILY or according to the manufacturer's directions.

Effective Management of Surgical Instruments

CANNULATED INSTRUMENTS (CONT.)

- Flush lumens using a syringe filled with warm distilled or deionized water according to the manufacturer's directions.
- Flush below the surface of the water to avoid aerosolizing contaminant

Effective Management of Surgical Instrument

ANNULATED INSTRUMENTS (CONT.)

- Follow rinse with air dry.
- 50cc syringe filled with air.
- Compressed air

Effective Management of Surgical Instruments

AUTOMATED CLEANING / MECHANICAL WASHING

- A washer-decontaminator can be used to mechanically wash sturdier instruments in an agitated detergent bath.
- Use detergent with NEUTRAL PH.

- Follow the recommendations of both the manufacturer of the detergent and the manufacturer of the instrument.

Effective Management of Surgical Instrument

AUTOMATED CLEANING / MECHANICAL WASHING (CONT.)

- Place heavy instruments in the bottom of the tray or basket.
- Load small or light instruments in separate trays.

Effective Management of Surgical Instruments

AUTOMATED CLEANING / MECHANICAL WASHING (CONT.)

- Do not put microsurgical instruments into a washer decontaminator unless it has a DELICATE CYCLE.
- Do not put powered or lensed instruments into a washer decontaminator unless recommended by the instrument manufacturer.
- Do not use a washer decontaminator for heat, moisture, or pressure-sensitive instruments.

Effective Management of Surgical Instruments

MECHANICAL CLEANING WITH HARD WATER

- Mineral deposits will accumulate on the walls of equipment after repeated use.
- Clean equipment regularly to remove impurities and scale.
- Water purifying system or softening agent added before each use minimizes formation of scum, scale, and discoloration.

Effective Management of Surgical Instruments

PREPARE THE INSTRUMENTS

- Just as in the OR, SPD personnel must insure that
 - Box locks are open
 - All multipart instruments are disassembled
 - All instruments in a set are together
 - All surfaces are exposed so that disinfectants and sterilants can reach ALL surfaces

Effective Management of Surgical Instruments

PREPARE THE INSTRUMENTS (CONT.)

- Protect sharp instruments
 - Sharp tips and sharp edges are easily dulled or broken.
- Protect personnel from injury from sharp instruments.
 - Don't bury sharp instruments among other instruments.
 - They should be readily visible to avoid injury when handling.

Effective Management of Surgical Instruments

ULTRASONIC CLEANING

- Ultrasonic cleaners use high frequency sound waves to create mechanical vibration.
- Cavitation: microscopic bubbles form on every surface, then implode, creating a vacuum which pulls particles from every crevice of the instrument.
- Ultrasound penetrates areas that a brush or mechanical washer cannot reach.

Effective Management of Surgical Instruments

- Use detergent at the proper concentration and temperature, recommended by the manufacturer.
- Ultrasound for 5 minutes or according to the manufacturer's directions.
- The ultrasound machine should be drained and cleaned/disinfected frequently.
 - Frequency of cleaning is based on the amount of use.
 - Frequency can be as often as every 1-2 hours or after every use.

Effective Management of Surgical Instruments

- Sort instruments into batches, separated by metal composition.
- Never place different metals in ultrasonic cleaner at the same time.
- Stainless steel and titanium instruments must be cleaned separately.
- Mixing incompatible metals can CAUSE PITTING, ETCHING, AND DEGRADATION of surface treatments.

Effective Management of Surgical Instruments

ULTRASONIC CLEANING (CONT.)

- Powered instruments should be not cleaned in an ultrasonic washer.
- A NEUTRAL PH ultrasound solution is also vital to protect the passivation layer on instruments and prevent corrosion.

Effective Management of Surgical Instruments

ULTRASONIC CLEANING (CONT.)

- PROTECT tips, cutting edges and box lock from other instruments.
- Ultrasonic vibration can cause premature wear when instruments contact adjacent instruments.
- Rinse instruments thoroughly with DISTILL WATER after the cycle to remove residue and particles from surfaces and to prevent staining

Effective Management of Surgical Instrument

ULTRASONIC CLEANING (CONT.)

- CHECK SCREWS on instruments, as vibration may loosen screws.
- Instruments should be AIR DRIED with filtered, compressed air.
- USE HOT AIR DRYER.
- Avoid canned, compressed air due to particulates.
- Protect fine tips and edges which can be damaged when snagged on gauze or towel.
- If drying with a towel, the towel should be LINT-FREE

Effective Management of Surgical Instruments

- Lubrication helps prevent staining, rusting and corrosion (if specified by the instrument manufacturer).
- Lubrication promotes smooth action of hinged instruments
- Proper lubrication helps to keep instruments clean by preventing mineral and protein build up.

Effective Management of Surgical Instruments

- Lubricate only instruments with mechanical parts.
- Use a water-soluble, anti-microbial lubricant after each cleaning, or a minimum of once each day.
- Water soluble lubricant penetrates the box locks, hinges and crevices, preventing binding and excessive wear.
- Do not lubricate cannulated instrument

Effective Management of Surgical Instruments

- Follow the manufacturer's instructions for lubrication.
 - Place instruments individually in a perforated tray; do not allow instruments to touch one another. Immerse the open perforated tray in the lubricant.
 - 30-40 seconds is the usual immersion time.
- Replace the lubricant at least daily to keep it clean.
- DO NOT USE SILICONE SPRAYS or other oils which can build up in hinges and crevices and inhibit sterilization.

Effective Management of Surgical Instruments

- Let the instruments drain.
- Do not rinse or wipe the instruments.
- Some lubricant will be removed during ultrasonic cleaning.

Effective Management of Surgical Instruments

- Lubricant film should remain on instruments through the sterilization process.
 - Though much of the lubricant is removed during preparation and sterilization, the lubricant that remains provides surface protection

Effective Management of Surgical Instruments

INSPECTING AND TESTING

- Magnified visual and mechanical inspection is a CRUCIAL STEP in the processing of instruments.
- Each instrument is inspected following the lubricant bath.
 - Misalignment, malfunction, dull edges or points on sharp instruments, bent tips, loose screws, pits, nicks, cracks, corrosion, etc.

Effective Management of Surgical Instruments

INSPECTING AND TESTING (CONT.)

- Damaged instruments are SET ASIDE for further evaluation, repair, or replacement.
 - Damaged instruments must never be put into sets.
 - A set should not be processed until the damaged instrument has been replaced;
- INCOMPLETE

SETS CAN CAUSE DELAYS IN SURGERY

Effective Management of Surgical Instruments

INSPECTING AND TESTING (CONT.)

- The instrument finish should have no dull spots, dents, nicks, or cracks.
- Incompletely cleaned instruments must be reprocessed.

Effective Management of Surgical Instruments

INSPECTING AND TESTING (CONT.)

- Sharp instruments (scissors, rongeurs, osteotomes, curettes, knives, etc.) should be tested for sharpness and smooth cutting under magnification.

- Hinged instruments should work smoothly without stiffness.

Effective Management of Surgical Instruments

FORCEPS

- Jaws that overlap when tightly closed are out of alignment.
- Teeth of forceps must mesh properly.
- Tying platforms must meet evenly over the full length to eliminate damage to fine sutures.

Effective Management of Surgical Instruments

- Hold the ring handles in both hands, open the instrument, wiggle the instrument to identify excessive play in the box lock.
 - Excessive play indicates alignment problems or wear.
 - Instrument may not hold securely.

Effective Management of Surgical Instruments

RATCHET

- TEST TENSION by closing the instrument gently; when jaws touch, a space of 1/16" or 1/8" should occur between the ratchet teeth of each shank.
- Close ratchet to first tooth.
- Hold instrument by jaws and tap against your hand.
- If instrument springs open, it is an indication of defective ratchet teeth or poor shank tension.
 - The instrument will not hold tissue, sponges, or needles securely

Effective Management of Surgical Instruments

SCISSORS

- When closed, scissor should meet only at tips.
- All scissors should operate smoothly.
- Rough operation implies wear, damage, or failure.
- Look for burrs on blade tips.
- Scissors with nicks in the blades should be identified and tagged for repair.
- Shanks should be in good alignment; if not, the blades may not close smoothly.

Effective Management of Surgical Instruments

- Many facilities use Theraband (the same latex product that is used for exercise bands) to test the sharpness of Mayo and Metzenbaum scissors.
- There are two thicknesses; one for scissors with blades longer than 5 inches and one for scissors with blades less than 5 inches

Effective Management of Surgical Instruments

NEEDLE HOLDERS

- A needle should be held firmly in place when the needle holder is closed to the lock or stop.
- If the needle can be twisted or rotated, the needle holder needs repair.

- If light can be observed between jaws that are closed and/or locked, the jaws are sprung or damaged and will require repair

Effective Management of Surgical Instruments

RONGEURS

- The cutting edges of a rongeur must be SHARP.
- A sharp Kerrison rongeur should take a clean bite out of a 3x5 index card.
- Bone-cutting rongeurs should bite through a tongue depressor.
- There are commercially available sharpness test kits and standards for surgical instruments.

Effective Management of Surgical Instrument

- Instruments are assembled and placed into containers for sterilization.
- All surfaces of each instrument must be exposed to sterilant.

Learning outcome 2: Perform Perioperative Theatre Technical services

Introduction to the learning outcome

Upon completion of this unit describes the knowledge, skills and attitudes required to adjustment of operating table, anesthetic machines and their operation, Suction machine and its operation, Diathermy machine and its operation, Lighting in operation theatre, Endoscopy/colonoscopy machine and their operation, Scopes and their operation and Operation Theatre accessories and appliances and their use

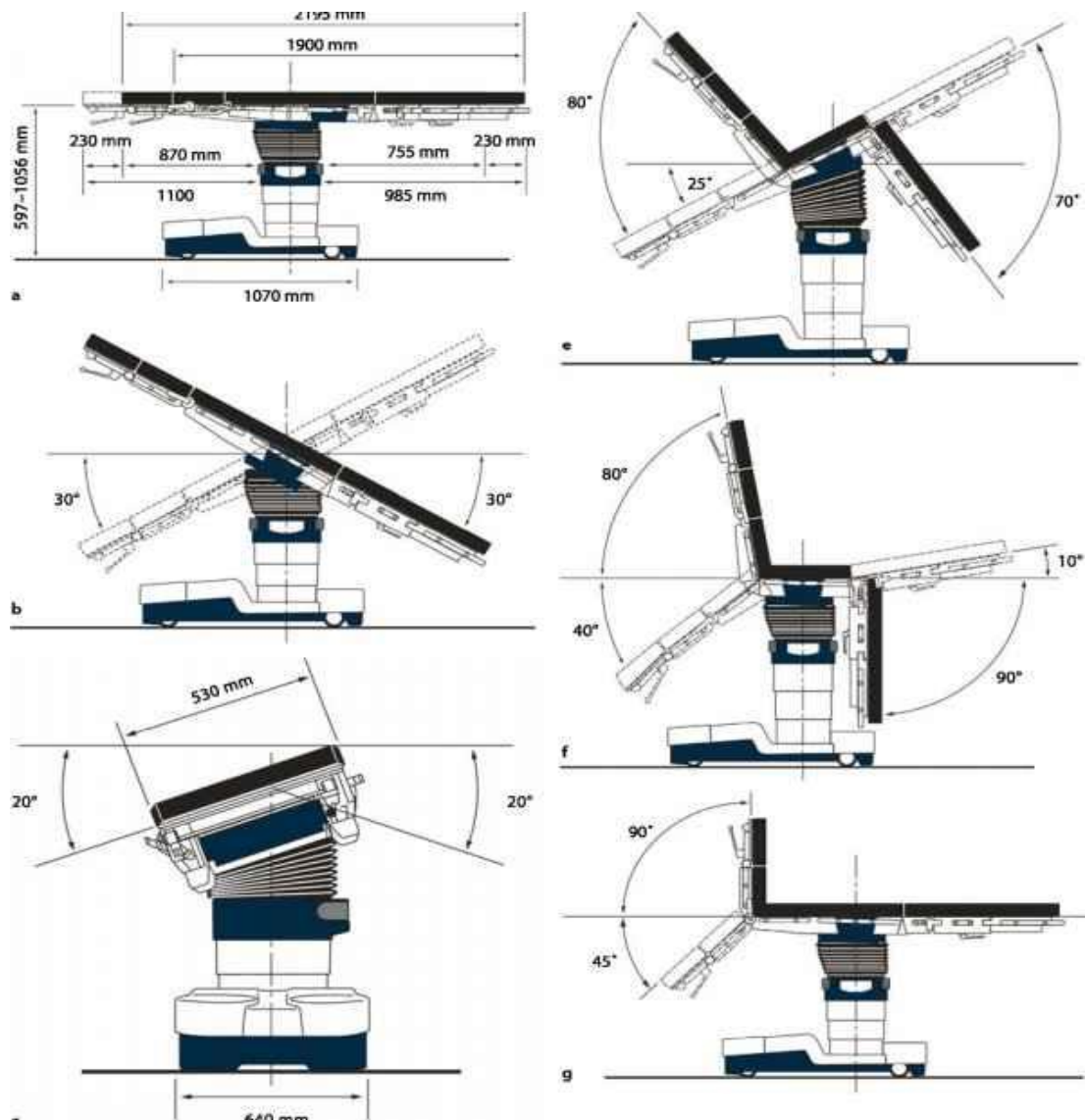
Performance Standard

1. Operation table is adjusted according to the procedure to be carried out.
2. Anaesthetic machine is operated as per user manual
3. Suction machine is connected as per user manual

4. Diathermy machine is set as per user manual
5. Operation lights are set as per procedure to be done
6. Accessories and appliances are set as per the procedure to be carried out

Information sheet

Adjustment of operating table



The electro hydraulic operating table that has a lift capacity of 1,000 pounds. In fact, this operating table can be lowered to just 24 inches off the floor in order to accommodate surgeons who are seated during lengthy operations.

Additionally, the table can be equipped with a 180 degree rotating table top for patient imaging access and offers:

1. Chest plate adjustment
2. Back plate adjustment
3. Head plate adjustment
4. Leg plate adjustment
5. Longitudinal shift
6. Other flex positioning

Anaesthetic machines and their operation



Describe the preparation of an anesthetic machine and anesthetic for a patient with known malignant hypothermia susceptibility

Clean the anesthetic machine; remove vaporizers; and replace CO₂ absorbant, bellows, and gas hose. Flush the machine for 20 minutes with 10 L/min of oxygen. Have the MH cart in the operating room? Schedule the patient for the first case of the day and notify the postanesthetic care unit to be prepared with an appropriate number of personnel. A cooling blanket should be placed under the patient. Refrigerated saline should be available.

A no triggering anesthetic technique such as continuous intravenous infusion of propofol should be used. The patient should be monitored for 6 to 8 hours after surgery.

Suction machine and its operation

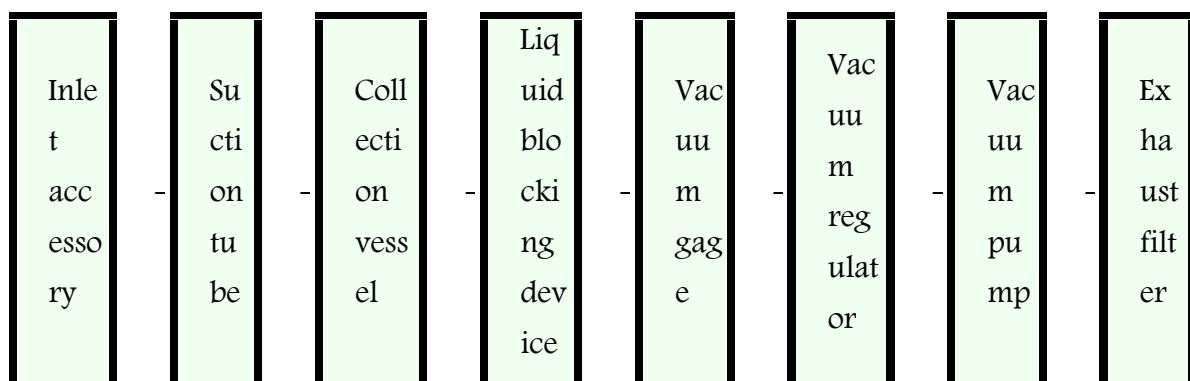
In surgery, suction is used to remove blood, saliva, other biological material and foreign detritus from an anatomical passage or from a working field. A suction-irrigator can simultaneously perfuse and collect a saline solution where washing is required.

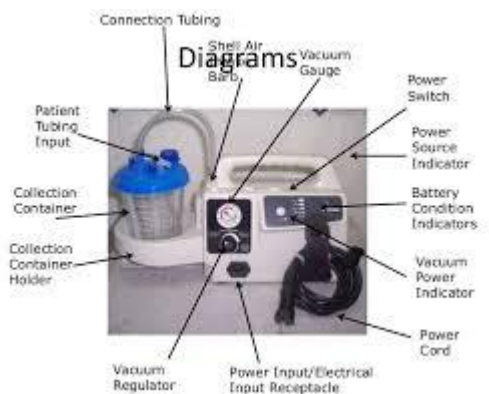
Suction is indispensable to clear the airway of blood, mucous, vomitus or other obstructing material. Consequently, suction machines can be found in emergency rooms, ambulances, patient wards and cafeterias.

A central vacuum pump connected to a building-wide network of suction plumbing is termed a "house vacuum". This is an efficient alternative to multiple small machines distributed throughout the building. Nevertheless the individual machines are necessary for mobile use and for backup during failure of a house vacuum.

Fluid Flow Path

filter Fluid Flow Path[[edit](#)]





Disinfecting Your Portable Suction Unit

It is best to follow the manufacturer's guidelines when disinfecting your suction unit, but in general, the following steps apply:

Disconnect the unit from its power source. Disconnect the battery from the PC board when cleaning the interior chassis.

Discard all disposable parts, including the canister, tubing, and catheters. Dispose biohazardous materials appropriately.

Use a mild detergent or a mixture of bleach and water (1-part bleach/ 10 parts water) and rinse thoroughly.

Follow the instruction manual when disinfecting the mechanics of the unit.

Never submerge the suction unit. Use disinfectant wipes to clean all outer surfaces, including control knobs, screens and handles.

Don't wait for the end of your shift to disinfect the suction unit. Do it immediately after each call to ensure it is clean and operational for the next patient.

Don't Forget About Power

Keeping your suction unit functional means ensuring it always has a reliable power source. The unit is of no use if the batteries are dead. Here are our recommendations:

Check the batteries at the start of each shift.

Test the batteries periodically, per manufacturer's guidelines.

Replace worn or damaged batteries.

Have plenty of backups on hand, especially for MCIs.

Practice Makes Perfect

Proper maintenance is important, but so is proper training. Here are some ideas for honing your suction skills:

Incorporate suction technique into code practice scenarios.

Review airway anatomy to maximize suction technique. Know the key differences between paediatric suction techniques.

Review the dangers and complications associated with improper suctioning.

Be aware of variations in suction catheter size/style and their appropriateness per patient.

Keeping the Unit Safe

Finally, to ensure your portable suction unit is ready for the next airway emergency, it is important to protect it from damage. This includes:

Avoiding extreme temperatures Keeping the suction unit away from wet or high moisture areas

Securing the unit within your rig to prevent damage while responding Keeping the unit free of dust and debris

Knowing how to properly clean your portable suction unit is only part of maintaining its overall health. Practice these recommendations to maximize your unit's efficiency and effectiveness.

Diathermy machine and its operation

Diathermy is the use of high frequency alternate polarity radio-wave electrical current to cut or coagulate tissue during surgery. It allows for precise incisions to be made with limited blood loss and is now used in nearly all surgical disciplines.

Diathermy uses high-frequency electric current to produce heat deep inside a targeted tissue. ... Instead, the waves generated by the machine allow the body to generate heat from within the targeted tissue. Diathermy is usually part of a complete physical therapy or rehabilitative regimen. Diathermy is the controlled production of "deep heating" beneath the skin in the subcutaneous tissues, deep muscles and joints for therapeutic purposes. There are basically two types of diathermy devices on the market today: radio or high frequency and microwave

Microwave diathermy uses electromagnetic radiation by microwaves and heats to a lesser tissue depth than short-wave diathermy. It is primarily used to heat superficial muscles and joints such as the shoulder

There is one basic difference between bipolar and monopolar techniques. With monopolar electrosurgery, a probe electrode is used to apply the electrosurgical energy to the target tissue to achieve the desired surgical effect. ... With the bipolar electrosurgical method a bipolar device, often a set of forceps, is used.

Surgical diathermy machine

In surgical diathermy, a high-frequency electric current (0.5—2 MHz) is used to produce heat to seal (by coagulation) blood vessels, or to cut and seal at the same time. The heating can be regulated by a variable resistance.

There are two types of diathermy machine: monopolar and bipolar.

In monopolar machines, the current flows from the active electrode through body tissue, along the line of least resistance, to a large indifferent plate electrode and

MAINTENANCE AND REPAIR OF LABORATORY AND HOSPITAL EQUIPMENT

Different waveforms can be used to produce a cutting or coagulating mode. For the cutting mode, the waveform is continuous, while for the coagulating mode, the energy is produced in bursts.

With bipolar diathermy, the two tips of a pair of forceps are the two electrodes; there is no plate electrode. Coagulation occurs when the forceps are closed across a piece of tissue. This technique is very good for dealing with very small blood vessels. The energy output required is low.

Diathermy machines can be classified according to the type of output connection used.

Isolated output machines have both sides of the output floating with respect to

earth. This minimizes the risk of possible alternative pathways for radiofrequency current (which can cause burns), or for mains frequency current from the machine itself (which can electrocute).

Radiofrequency earthed output machines have a capacitor placed between the plate side of the output and earth. This permits radiofrequency current to flow to earth. The small value of the capacitor prevents the passage of mains frequency current so that isolation is effectively preserved at the mains frequency; the risks of electrocution are not increased by this method. However, the risk of possible alternative pathways for radiofrequency current to cause burns is greatly increased and for this reason a plate current monitor is required on such machines. This monitor senses any difference in current flows between the active and the plate leads, and turns off the output if a difference occurs. The difference represents the leakage current.

Hard-earthed output machines have a plate electrode directly connected to earth, so, the patient is also directly connected to earth. This means that the risk of electrocution is very much higher. For this reason, machines of this design are no longer produced, and any that are still in use should be replaced.

Maintenance

The routine maintenance needed for these machines is limited. Proper records should be kept of all work done on them. Make a check-sheet of work to be done,

and then make routine inspections at least twice a year. The most important job is the electrical safety check. If you do not have an electrical safety checker, follow the instructions in the section on cardiac monitors

Keep the inside and the outside of the machine clean. Check the plate electrodes, the leads, and the instruments. Most of the problems will be caused by faulty leads, and in many cases, you should be able to repair these yourself. The instruments (e.g., diathermy forceps) are unlikely to give problems, provided that they are checked after each sterilization.



- Lighting in operation theatre



Operating lights are a necessary component of any operating theatre or surgical suite and are available in a variety of different configurations, wattage amounts and features. Avante

carries a wide selection of surgery lights including both large, ceiling-mounted operating room lights and small, portable exam lights. In recent years, older halogen lights have been replaced by brighter, more energy-efficient LED options. LED surgery lights last longer than traditional incandescent versions and often incorporate LCD touchscreens to adjust illumination and spotlight settings.



Repairs, maintenance and medical equipment planning is best delivered by a company who specializes in the up time of operating room service and support within surgical centres. Operating rooms are specifically designed for surgeons and surgical staff to perform surgical procedures that require time, patience, focus, and safety. Various pieces of equipment are required for use in the operating room. Let ERD be your professional medical equipment planner, they are vendor neutral.

Surgical Lights – High-quality surgical lighting is essential for performing intricate procedures in the operating room. Surgical lights are designed to provide bright white light to illuminate the surgical site and eliminate shadows, all while keeping the surgical staff cool. LED lights or halogen lights are two types of surgical lights. Surgical lighting is transitioning away from incandescent technology to LED technology due to the many benefits of LED: pure white colour, less heat in the surgical field, more accurate colour rendition, and improved shadow control. Recommended maintenance is 2x per year.



- Endoscopy/colonoscopy machine and their operation

What is endoscopy?

Endoscopy is a procedure in which the gastro-intestinal tract (GI tract) is viewed through a lighted, flexible tube with a camera at the end (endoscope). Small samples of tissues cells (biopsy) can also be collected and sent for testing.



In Procedure rooms a clean to dirty workflow must be maintained. The clean area will include

sterile supplies and a write-up space with computer and printer to generate endoscopy reports.

The room may be purpose designed to accommodate the following:

endoscopy 'stack' and video monitor(s) – this equipment contains the light source and video processor required for the endoscopes to produce images

endoscope cabinet with clean endoscopes and accessory equipment such as endoscopy biopsy forceps, snares, injectors

monitoring equipment to allow continuous monitoring of patient condition during procedures

anaesthetic equipment and medication used to provide procedural sedation or short acting anaesthetics, diathermy and/or Argon plasma coagulation equipment

imaging equipment such as Image Intensifier or C- arm X-ray screening unit depending on procedures to be performed; imaging equipment may be portable or installed in the room.

Endoscopy Unit

International

Health Facility Guidelines

scope is wiped down and placed in a closed container and transported to the clean-up area; if any

delay in processing is expected the scopes are soaked in an enzyme detergent solution.

Decontamination includes leak testing, manual pre-cleaning followed by high level disinfection with

a disinfectant solution.

Endoscope reprocessing areas should be separate to Procedure/ Operating rooms and unidirectional dirty to clean workflow must be maintained. A centralised reprocessing area is recommended for efficient handling of endoscopes and appropriate air pressurisation and ventilation. Cleaning and Disinfection areas must be negatively pressured and ventilated to remove vapours of chemicals used in the process.

The Cleaning and Disinfection of equipment area will include:

sinks for soaking and rinsing sufficiently sized to prevent tight coiling of the endoscope which may damage the fibre-optic cables in the instrument

ultrasonic cleaner for accessory equipment used in procedures

Automated endoscope cleaning/ disinfecting machines

compressed air to aid drying of endoscopic equipment after cleaning

handwashing basin

safety eyewash facility

stainless steel benches with space to accommodate the length of the endoscopes

storage for disinfected scopes on a bench or shelf.

The Sterilising/Disinfection area will include an autoclave to sterilise accessory instruments if a

sterile supply service is not available. Storage provisions will include adequately sized areas for drugs, sterile stock, consumables, linen,

resuscitation trolley and mobile equipment which may include mobile imaging equipment. Storage of sterile items and scopes close to the point of use is recommended. Scope storage areas must

be positively pressured and HEPA filtered to prevent contamination of clean endoscopes. Scopes

may be stored in properly ventilated and temperature-controlled cabinets, preferably a passthrough type, located between reprocessing/ sterilising/disinfection areas and the Operating/

Procedure room. Endoscope cabinets should allow for endoscopes to hang without coiling preventing damage to either end of the scope.

- Scopes and their operation

Surgical scope



An operating or surgical microscope is an optical instrument that provides the surgeon with a stereoscopic, high quality magnified and illuminated image of the small structures in the surgical area.

The optical components of a basic stereo microscope consist of the binocular head, a magnification changer, the objective lens and an illuminator which beams light through the objective lens and onto the operating field (Figures 1 and 2). The binocular head consists of two telescopes with adjustable eyepieces for users with refractive error. The magnification can be changed by turning a knob (which selects different magnification lenses) or by using a motorised zoom controlled by a foot pedal.

The working distance (Figure 1) is the distance from the microscope objective lens to the point of focus of the optical system. This value is fixed and is dependent on the chosen focal length of the objective lens. The choice of working distance depends on the type of surgery. For modern ophthalmic surgery that involves delicate work in the posterior chamber, objective focal lengths of 150 mm, 175 mm and 200 mm are commonly used.

The optical system often includes a beam splitter and a second set of teaching binoculars (Figure 2) so that two people can view the operation simultaneously.

The optical system is attached to the suspension arm of the floor stand (Figure 3). The suspension arm makes it possible to position the optics exactly and to fix them in place.

The floor stand has wheels and can be moved around the floor and fixed into place using the brakes.

A foot pedal connected to the floor stand allows the surgeon to control the focus, the zoom, the position of the optics over the eye (the x,y position on the horizontal plane) and to turn the illumination on and off.

The illumination system is usually housed in the floor stand in order to keep the bulb heat away from the operating field. In this case, the light is transmitted to the operating field by means of a fibre optic cable. The light in ophthalmic microscopes is usually coaxial, meaning that it follows the same path as the image in order to avoid shadows.

Procedures requiring scopes includes

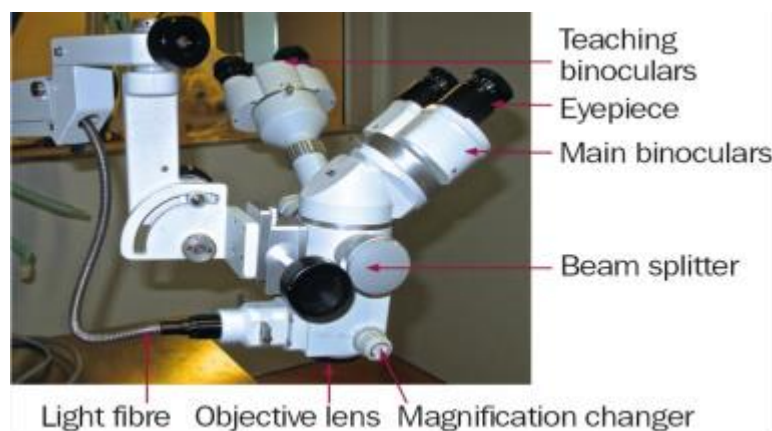
Neurosurgery and Spine

Ophthalmology

Otolaryngology (ENT)

Dental

Plastic/Reconstructive Surgery



The surgical microscopes of Leica Microsystems are exactly geared to the requirements of microsurgery. A compact optical unit delivers clear and sharply focused images and the modular system gives the surgeon optimum maneuverability

Safety considerations When used in ophthalmology: • Avoid looking directly into the light source, e.g. into the microscope objective lens or a light guide. Any kind of radiation has a detrimental effect on biological tissue. This also applies to the light illuminating the surgical field. • Reduce the brightness and duration of illumination on the surgical field to the absolute minimum required. • When operating on the eye, always use a protection filter to ensure that the patient's retina is not exposed to unnecessary (blue) radiation (retinal injury). dr. Chris R. Mol, BME, NORTEC, 2015 Operating microscope • Stability of stand must be ensured • The system must be connected to a special emergency backup line supply (in case of power failure)

- Operation Theatre accessories and appliances and their use

operating room equipment list for operation theatre

By china care medical | 02 September 2018 | 11 Comments

Are you looking for operating theatre design guidelines for your new clinic or hospital operating room?

Do you want to get a operating room equipment list for operating room preparation for surgery?

Do you know the difference between cardiac operation theatre equipment, orthopaedic operating room, emergency operating room and other operating room equipment ?





, Surgical Lamp Light

It has the wall mounted and Floor stand type, it shows the wall mounted type in the above picture.

According to the light source principle. it has the LED, Apertured bulb type, the LED types is always with longer working life, and brightness.

Bulb type is more cost efficient.

More other types with price, please click [her2](#), x-ray machine

it is a kind of surgery x-ray machine, 2-1 is the control of the x-ray machine, 2-2 is the main unit of the x-ray machine.

There are different types x-ray machine, from the portable type to stationary type.

3, Surgical table bed

It is used for the patient to sit and lie, it has different types, electric and manual types and for different surgery type, it has different surgery table for different usage.

4, patient monitor

it is used for monitoring the patient conditions during surgery.

for adult and pediatric and pregnant patient, there are different types, also different machine can monitor for different parameters, and different size screen of the machine,

5, Trolley

it is used for putting medicine, instruments and other useful items,

different size and different styles also different material will be influenced the price,



6, Anaesthesia machine

it is used for anaesthesia patient during the surgery,

7, Infusion pump

it is used for infusion patient automated even when patient is without awareness. some models can be with drug input functions,

8, Pendant

It is used for put patient monitor, endoscope and other surgery items during the surgery.

there is also other commonly used equipment used for surgery room.

Suction machine,

Blood Warmer,

Defibrillator,

ECG Machine,

Electrosurgical Unit,

Loupe,

Oxygen Generator,

Suction Machine,

Surgery Saw and Drills,

Ventilator machine,

Learning outcome 3: Evaluate Perioperative Theatre Technical services

Introduction to the learning outcome

Upon completion of this unit describes the knowledge, skills and attitudes required to Operation Theatre maintenance procedures, Referral of faulty equipment and apparatus and Monitoring and evaluation in Operation Theatre.

Performance Standard

1. Faults committed are identified and rectified or referred as per workplace procedure
2. Faulty equipment and apparatus are identified and reported to appropriate personnel

Definition of terms

Quality improvement

The OT nurse needs to have enough knowledge on the various instruments for performing risk assessments. To be able to view quality work from an overall perspective, the OT nurse needs knowledge of quality improvement (QI) on an organizational and health system level, such as set goals for the care of the intraoperative patient and guidelines from different professional organizations (Spruce, 2015).

The aim of QI projects is to determine whether the appropriate standards of care are reached in a specific clinical setting and in relation to an acceptable standard (Baker et al., 2014). The focus is placed on systems, at local and organizational levels and not on individuals. It is an internal procedure to cyclically evaluate work processes, which include EBP, clinical effectiveness, evidence-based clinical guidelines and audit (Rycroft-Malone et al., 2002). QI projects usually do not have theoretical underpinnings; they most commonly do not aim to generate new knowledge and are not generalizable to all settings (Baker, 2017; Cepero, 2011; Raines, 2012).

The foundation of nursing practice has three components that are closely related to each other, but with different purposes, QI, EBP and nursing research. QI aims to improve processes, EBP aims to change practice and nursing research aims to generate new knowledge (Hedges, 2006). To apply QI, knowledge concerning implementation is essential, which is defined as “methods to promote the systematic uptake of research findings and other EBPs into routine practice and, hence, to improve the quality and effectiveness of health services” (Eccles & Mittmann, 2006, p. 1). There are several different approaches in implementation science (Nilsen, 2015), whereof i-PARIHS (implementation—Promoting Action on Research Implementation in Health Services) is one of the most common in nursing to determine how the improvements can be detected and measured (Harvey & Kitson, 2016). To evaluate QI, benchmarks are most often needed, to allow the results to be compared with other entities’ outcomes (Baker, 2017). OT nurses’ in-depth nursing care knowledge, academic skills and an ethical approach in the perioperative context is needed to enable systematic QI.

4.5 Safe care

The fundamentals in systematic safety work are to identify and reduce adverse events to minimize patient suffering and preventable injuries (Rutberg, Borgstedt-Risberg, Gustafson, & Unbeck, 2016). Surgical patients are exposed to several risks during surgery;

risk of surgical site infection due to a planned break in skin integrity and risk of physical injury due to surgical positioning, electricity, chemicals and transfers (Rauta, Salanterä, Nivalainen, & Junttila, 2013). A large proportion of these surgical complications are avoidable, which was shown when the World Health Organization (WHO) surgical safety checklist was implemented, after which mortality was almost halved and complications were significantly reduced. The OT nurse is responsible for and verbally confirms, sterility, equipment availability, results of needle, sponge and instruments count and the handling of specimens in the WHO checklist (Haynes et al., 2009).

There are several areas for which the OT nurse possesses special knowledge and skills and therefore has special responsibility. Such an area is the prevention of postoperative surgical site infections. The OT nurse guarantees a hygienic and aseptic environment throughout the operation by creating a safe and sterile working surface for the use of surgical instruments and by draping the patient, thus creating a barrier between the wound and surrounding microorganisms (Kilvered et al., 2012). Moreover, the OT nurse is also responsible for controlling the traffic flow in the OT (Blomberg et al., 2015). Door openings during surgery affect the air pressure in the OR, which may allow contaminated air to flow in and lead to unacceptable numbers of airborne bacteria-carrying particles and potentially result in surgical site infections (Andersson, Bergh, Karlsson, Eriksson, & Nilsson, 2012).

The collection of specimens is common during surgical procedures, aiming to verify or support the patient's diagnosis (Rauta et al., 2013). The OT nurse is responsible for handling these specimens; an error can endanger the possibility of diagnosing the patient and planning further medical treatment.

Surgical counts are another area within the OT nurse's responsibility (McGarvey, Chambers, & Boore, 2000). The OT nurse's signature in the digital operation notes is a guarantee that surgical counts and qualitative assessments of the surgical instruments have been performed before the operation starts, during the procedure before wound closure and after the operation to ensure that no material (retained foreign body) is accidentally left in the patient. Preventing retained surgical items is one of the highest priorities in patient safety issues identified by OT nurses (Steelman, Graling, & Perkhounkova, 2013).

- Operation Theatre maintenance procedures

OPERATING THEATRE PROCEDURES AND EQUIPMENT

Two things are essential for an operating theatre to run effectively and efficiently.

The first and most important is sterility. The purpose of all the precautions and care taken in operating theatres is to prevent infection occurring at the time of operation. This is particularly important for eye surgery where infection is not just a complication but a disaster. An additional risk in eye surgery is that intraocular damage may be caused by chemicals or inappropriate irrigating solutions entering the eye.

The second is teamwork. Surgery is not the work of one important person, “the surgeon”, with a few other people who are not so important doing what they are told. It is the work of a whole team. Everybody in the team is equally important although obviously the surgeon has had a longer training than the others.

The old proverb “the strength of a chain is its weakest link” is particularly true for operating theatre staff and procedures (see fig. 3.1). It only needs one dirty instrument to introduce infection to the eye and destroy it. It only needs one person in theatre to make a mistake with fluids which are irrigated in the eye, to cause destruction of the corneal endothelium and blindness. Everyone must share the responsibility for safety and sterility.

The reason for all good surgical practice is to make the operating theatre a safe

place.

1. Safe for the patient who must be protected from infection and other harm. The patient is the most important person who needs protection.)

2. Safe for the staff from the risk of needle-stick and other injuries. These can spread Hepatitis, HIV and other infections. In those parts of the world where hepatitis and HIV infections are common, “every used needle and sharp instrument is as dangerous as a loaded gun.”

3. Safe for the community by careful and safe disposal of soiled materials, especially sharp instruments and needles. In developed countries most eye surgery takes place in very sophisticated

operating theatres dedicated to eye surgery alone, with staff who are fully trained. In developing countries eye surgery often takes place in general theatres, and some takes place in buildings which may have other uses and are only temporary

operating theatres. Often the staff are not fully trained and accredited as nurses or

theatre technicians. In spite of these short-comings many surgical teams have to the chain of successful surgery – if any one link is broken, the surgery will not be successful cope with a large volume of work and often with very limited facilities. While safety and sterility are obviously the most important aspects of all theatre work, efficiency is also important because of the large volume of work.

Another aspect of safety in operating theatres concerns looking after unconscious patients receiving general anaesthesia, or resuscitating a patient who may have collapsed. Everyone working in a theatre team should know about “basic life support”. This is how to care for an unconscious patient or one who has collapsed.

Basic life support means how to maintain a clear airway and how to give artificial respiration to a patient who has stopped breathing, and how to check the pulse and give cardiac massage to a patient who has no heart beat. All theatre staff should regularly practise these skills, so as to be ready if and when an emergency occurs.

At least one member of the theatre team should be trained in “advanced life support”, which involves taking and interpreting ECG’s, the use of a defibrillator and giving appropriate intravenous fluids and drugs.

Theatre procedures depend on many different factors: the work-load; the equipment available; the choice of the surgeon; etc. Eye surgery is practised in many different ways especially in developing countries. There are some highly effective units with a massive through-put of cases. On the other hand, there are some surgeons who are only part-time ophthalmologists, and there are many mobile teams working in temporary accommodation. therefore, this chapter only discusses basic principles and guide-lines.

The Theatre Team It is important to have an adequate number of motivated and enthusiastic staff. If they are dedicated to eye surgery, they will have a better understanding of the requirements of both the surgeon and the patient. A happy relaxed atmosphere and good working relationships in the theatre team makes the work much more pleasant and mistakes less likely.

The theatre team should have a leader who takes responsibility for organising the work and making sure all the routine jobs are done regularly. This ensures both the safety and efficiency of theatre work. For a regular operating list the basic personnel required are:-

1. A scrubbed assistant to look after the sterile instruments and assist the surgeon.
2. An anaesthetic assistant to give the local anaesthetics, prepare the patients and if required help with any general anaesthetics.
3. A circulating nurse to clean and sterilise the instruments and apply the dressings.
4. A general assistant.

It is often appropriate that the team leader assumes a fairly basic position, such as being the general assistant, in this way he or she can supervise all the other team members.

Eye Surgery in Hot Climates

Operating Theatre Routine

There are a variety of important jobs that must be done to keep a theatre well

stocked and maintained. Many of these are rather obvious, but without maintenance equipment will break down, and without planning spare parts and consumable materials will run out and take a long time to be replaced.

The general routine will include:

1. Building maintenance.
2. Cleaning.
3. Maintenance of equipment and instruments.
4. Manufacture of dressings and drapes.
5. Sterilisation and disinfection procedures.
6. Stock-keeping, storage and security.

1. Building maintenance

A good sound building is an obvious requirement for safe surgery. Eye surgery can be performed in a great variety of buildings which are not purpose-built operating theatres.

However, the room should be as insect-proof as possible and well ventilated. It does not have to be blacked-out, although the windows should be shaded. Paint work should be in good condition and a secure water supply present.

The room should have doors that can be closed during surgery. Regular inspections of the insect-proofing are important.

2. Cleaning

General cleaning should be carried out regularly in addition to preparations on the day of surgery. Floors and sometimes walls and ceilings must be washed in all

rooms used as part of the operating theatre suite. Any furniture including instrument tables, operating tables and cabinets must be wiped clean to avoid the build

up of dust.

Spilt blood or other debris should be wiped up as soon as possible, because once dried it may be difficult to remove. A weak solution of bleach is adequate for cleaning purposes and will kill most micro-organisms including the HIV virus.

Anyone who washes drapes and surgical instruments MUST wear gloves to protect themselves from the risk of infection.

3. Maintenance of equipment and instruments

Equipment can only function if it is regularly maintained. A schedule needs to be drawn up for items such as sterilisers, operating lights and air conditioners. The importance of having spares to enable quick repairs to be carried out locally cannot be stressed too much. Surgical instruments need to be carefully looked after and checked that they are working properly.

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4. Manufacture of dressings and drapes

Eye pads.

With modern surgery and small self-sealing incisions, patients are no longer routinely given an eye pad postoperatively. However, many patients still require an eye pad postoperatively and outpatients may also be padded as part of their treatment. The purpose of the eye pad is to protect the eye in the immediate post-operative period. It also prevents the patient or his attendants from interfering with the eye and keeps flies out. It prevents eye-lid movements against the eye and applies gentle pressure which will encourage haemostasis and wound closure.

An eye-pad is not a substitute for poor surgery, and it is rarely necessary to keep an eye padded for more than 2 days postoperatively.

The cost of buying ready-made pads is high and there are inevitable delays as a result of ordering supplies. Eye pads can be manufactured locally using cotton wool and gauze. A

layer of gauze is placed on a table and onto this is put a layer of cotton wool about 2 cm thick. A further layer of gauze goes on top of the cotton wool making a cotton wool sandwich enclosed in gauze. Then, using a simple card shape as a guide, eye pads can be cut out.

These are placed into an autoclave box or bin for sterilization. The technique for applying pads is simple but important. The pad needs to be placed so that the eye-lids will not be able to open. If the eyelids open under the pad, the pad will rub against the cornea and this will harm the eye and not protect it.

The pad is placed diagonally over the closed lids. If the eye is very deeply set it is best to fold the pad in half and place it so that the folded edge fits below the eyebrow. The pad is taped firmly in place using three strips of adhesive tape, ideally 1 cm wide and positioned.

Added protection can be provided by a plastic eye-shield. These can be bought or made locally from old X-ray plates or cardboard.

In most surgical cases it is safer to bandage the eye with elasticated or crepe Applying an eye pad How to make a simple eye shield from X-ray plates

5. Sterilisation and Disinfection Procedures

The sterilising of instruments, swabs, linen and eye-pads is by far the most important step in safe surgery. Sterilisation means that all living micro-organisms, bacteria, viruses, fungi etc. including spores have been killed. Disinfection means that bacteria which are likely to cause infection have been killed, but spores and some very resistant micro-organisms may survive disinfection. Obviously sterilisation is better than disinfection. There are four common ways of sterilising or disinfecting.

Methods of Sterilisation and Disinfection

1. Autoclave
2. Dry heat oven
3. Boiling

4. Immersion in chemical solutions

Autoclaving and the dry heat oven will sterilise, and boiling and chemical solutions

will only disinfect. However, the methods of sterilisation may only disinfect if the

treatment is not applied for long enough, and a chemical which disinfects may sterilise after a long period of immersion. Instruments must first be cleaned before sterilising. The best time for cleaning

instruments is immediately after they have been used, otherwise blood and secretions may become dried and encrusted. Dried blood and secretions are much more difficult to remove, and they prevent spores and bacteria from being killed by the sterilisation process. Instruments should be washed with soap and water using a soft brush or cloth, paying particular attention to the joints of scissors, artery forceps and needle holders. They should then be rinsed in clean water.

If instruments are going to be stored or sterilised using dry heat or chemicals, they should

also be carefully dried. Theatre linen and drapes can be washed at the end of the

list and left to dry in the sun.

1. Autoclave

The autoclave is by far the safest, best and most reliable way of sterilisation. It is the only effective way of sterilising surgical drapes and dressings, and is

the best way to sterilise surgical instruments. If used properly it is guaranteed to kill

both bacteria and spores. The autoclave produces water vapour at pressures in

excess of 1 atmosphere, and this enables temperatures up to 134 degrees Centigrade to be achieved. The higher the pressure, the higher is the temperature at

Operating Theatre Procedures and Equipment

A Heavy needle holder for large needles

B Fine needle holder for fine needles

C Conjunctival scissors

D Intraocular scissors (Vanna's)

E Right and left corneal scissors

F Iridectomy scissors (DeWecker's)

G Razor blade fragment holder

H Scalpel blade (Bard-Parker)

I Heavy toothed forceps

J Conjunctival forceps

K Fine corneo-scleral forceps with teeth at the tip

L Fine corneo-scleral forceps with cups at the tip

M Curved fine corneo-scleral forceps

N Intraocular forceps for handling the lens capsule (Kelman McPherson)

O Capsule forceps for intracapsular extraction (Arruga)

P Fine suture tying forceps

Q Right and left lens expressor

R Lens loop

S Iris repositor

T Hot point cautery for use with spirit lamp

U Irrigating cystitome

V Adjustable light speculum

X 2-way irrigating/aspirating cannula for extracapsular extraction

Y Intraocular swabs

Z 2 cc and 5 cc syringes

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A basic set of instruments for intraocular surgery

Operating Theatre Procedures and Equipment 47

A Needle holder

B Straight scissors

C Curved spring scissors

D Scalpel blade

E Squint hooks

F Fine straight artery forceps

G Curved artery forceps

H Conjunctival forceps without teeth

I Heavy toothed forceps

J Fine toothed forceps

K Battery cautery

L Meibomian cyst lid clamp

M Small curette

N Eyelid retractor (Desmarres)

O Eyelid guard

P Large eyelid clamp

Q Eyelid clamp (Cruikshank or Erhardt)

R Adjustable speculum (Laing)

48 Eye Surgery in Hot Climates

A basic set of instruments for extraocular surgery on the eyelids

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autoclaving has been completed satisfactorily. Special care should be taken when using an autoclave with attention to correct procedures and safety. The reservoir should be checked and kept at the right level. Only distilled or rain water should be used. Correct venting is essential although many autoclaves vent themselves at the end of the cycle. All staff using the autoclave require training and must strictly adhere to the instruction manual.

2. Dry Heat Oven

A hot air oven is quite expensive but requires very little maintenance and is a very effective way of sterilising instruments. It will preserve the edge of delicate cutting instruments better than an autoclave. However, instruments must be scrupulously clean and dry first, and the process is very slow. It takes 1 hour at 180 degrees Centigrade to sterilise instruments, and then the instruments must be left to cool.

3. Boiling

The great advantage of boiling is that it is very quick, simple, easy and cheap.

However, boiling does not sterilise, it only disinfects. Boiling for 10 minutes will kill all bacteria but it will not kill spores. Although this is a problem in theory it is not a real problem in practice. However, boiling has two particular disadvantages:

1. Repeated boiling will cause tarnishing and corrosion of instruments and will certainly, damage the cutting edge of fine instruments. Corrosion is lessened by adding 2% sodium carbonate to the solution. If possible, distilled water should be used and not tap water. This is available commercially from garages as water for car batteries. Distilled water will not cause corrosion.

2. After boiling the instruments are wet. They can be dried with a sterile gauze but this might compromise sterility and will break the “no touch technique” rule of not touching the instrument end that enters the eye. If they are used when wet, water from the handle of the instrument will drip down on to the working end and again, compromise the no touch “technique”. If the instruments are placed in the boiler in a perforated rack or tray, then they will dry automatically and not require handling. In emergencies, drapes can be sterilised by boiling and then squeezed dry, although this is certainly not a technique to be recommended.

4. Chemical Solutions

Most chemical solutions are effective against bacteria but not against spores, and so they only disinfect but do not sterilise. However, if the instruments are left in the

chemical for long enough they may indeed sterilise. Chemical sterilisation and disinfection is good at preserving sharp instruments but there are two main disadvantages.

1. These chemical disinfectants are very toxic especially to the inside of the eye, and the instruments must be thoroughly rinsed in sterile water and dried very carefully after immersion in the chemical. This again will break the “no touch technique” rule.
2. With time or use, the solution may lose some of its potency, or may become chemically inactivated, or the alcohol in alcohol based chemicals may evaporate.

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The most commonly used solutions are Chlorhexidine (trade name Hibitane) or Chlorhexidine and Cetrimide combined (trade name Savlon).

Hibitane is available as a 5% concentrate and needs to be diluted 10 times with 70% alcohol to make a solution which is 0.5% Chlorhexidine (Hibitane) in 70% alcohol and 30% water (i.e. one part of concentrate for 9 parts of 70% alcohol).

Savlon is available as a concentrate containing 1.5% Chlorhexidine and 15% Cetrimide and needs to be diluted thirty times in 70% alcohol (1 part of concentrate for 29 parts of 70% alcohol).

Both of these solutions should disinfect instruments after 10 minutes. After prolonged usage some of the alcohol may evaporate and the solution may lose its strength. If alcohol is not available concentrated Savlon diluted 1:30 in boiled

water will disinfect instruments after 30 minutes.

Povidone Iodine is another widely used disinfecting agent, which is effective against most micro -organisms. A 10% solution can be made as follows :

Take 500ml of distilled water (cooled, boiled rain water is an alternative)

Add sodium phosphate 16.6 gms and citric acid 3.4gms(these act as a chemical buffer)

Then add 50 gms of Povidone Iodine, thus making a 10% solution.This will also disinfect instruments after 10 minutes.

A 10% solution of Povidone Iodine can be used for cleaning the eyelids and the conjunctival sac preoperatively, but if it is left in the conjunctival sac for some time, some people think that 5% is safer.

Formalin vapour is an effective sterilising agent against all micro-organisms and spores. It is supplied as formalin tablets, which are placed in an airtight container at a warm room temperature. 12 hours are required for sterilisation.

The sterilisation of cryoprobes often presents a problem. They can only be autoclaved at lower temperatures and so will take half an hour for the cycle.

Cryoprobes are quite often sterilised by immersion in chemical solutions between patients on the same list. As the probe actually enters the eye great care must be taken both to wash off all these chemicals, and also to make sure that the probe is not in any way contaminated. A small portable cryoprobe used with freon gas can however be safely boiled. (Do not boil the cylinder of gas itself, as it will explode!)

Sterilised instruments, drapes and dressings can be safely stored and transported in sealed autoclave bins and trays. If such bins or trays are not available all

items must be double wrapped in linen before sterilisation and can then be safely stored or transported.

Stock keeping, Storage and Security

Good stock keeping to maintain essential supplies is often overlooked. However its importance cannot be over emphasised especially when there may be long delays between ordering supplies and their arrival. A system of monitoring stores and the rate at which consumables such as medicines, dressings, sutures etc. are used will allow for ordering and budgeting. Money is always limited so there is no place for holding excessive quantities. The only way to be aware of annual usage and any seasonal variations in consumption is to have a strict system of stock keeping. This

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important part of theatre management should be the responsibility of the person in charge of the theatre.

It is obvious that equipment and supplies should be stored in a place where they will not deteriorate, and where they are safe.

Managing with Limited Resources

In many circumstances the volume of surgical work or financial limitations mean that correct theatre procedures cannot be followed strictly. In correct theatre procedures everybody who enters the operating theatre and all the staff and patients should wear a complete change of clothes. All those operating scrub up

completely between cases with fresh gloves and a fresh gown. Often it may not be possible to maintain these standards. The patients may have to come to the operating theatre wearing their own clothes. The surgical staff should always scrub up completely at the beginning of a list, but may be obliged just to change their gloves in between cases or even to wipe their gloves in alcohol between cases. In order to maintain surgical throughput it may be necessary to have more than one operating table in the same operating room.

To ensure safe surgery there are four areas in which “cutting corners” and compromises are strictly forbidden. These are:

1. Correct preparation of the patient for surgery, with a thorough cleaning of the face and skin preparation around the eye (see pages 57–8). This includes “marking” the eye for surgery to avoid mistakes.
2. Sterilisation of all instruments, drapes and dressings. If an item like an intraocular lens comes from the manufacturer in a sterile wrapping, then make sure that the packaging has not in any way been damaged.
3. Sterilisation and purity of all solutions used in intraocular irrigation.
4. The correct handling of instruments and dressings and the use of a “no touch technique”. Nothing that enters the eye should touch the eyelids or lashes.

People working in small units, or on tour in rural areas may have to rely on boiling and chemicals as the only means of disinfecting between cases. Although this is not the best way it appears to be safe as long as everyone sticks to the rules. Some

people will boil the blunt instruments and use chemicals for the sharps.

- Don't try to shorten the time needed for immersion in chemicals or boiling.
- Make sure that the chemical solution is fresh.
- Make sure that the instruments are dried with a sterile swab and there is no chemical residue on the instruments. This should be done by someone who understands how important this is.
- Never rely on just wiping the instruments between cases with spirit, ether or acetone. It just isn't safe enough.

52 Eye Surgery in Hot Climates

Infection Control Strategy

Post-operative infections do sometimes occur, even with the most careful and well trained surgical team. Good units have an infection rate of about one case in a thousand or less. It is reasonable to assume that any infection developing within the first post-operative week has been contracted at the time of the operation. Even an isolated case of post-operative infection should make the surgeon and the surgical team review all their techniques, equipment and procedures. If several infections occur close to each other an even more radical overhaul of theatre procedures is required. Any irrigating fluids should be discarded and a new batch obtained. All made up disinfectant solutions should be discarded and new ones made, and the steriliser changed or a new method of sterilisation tried. If available an infection control officer or a microbiology department will help to trace the

source of an infection. The following is a list of the more common possible sources of infection during surgery:

Sources of Infection

From the patient Conjunctivitis.

Blepharitis.

Dacryocystitis.

Septic lesion near the eye.

From the staff Septic lesions.

Poor scrubbing up.

Contaminated gloves.

Failure to observe "no touch technique" by the surgeon or assistant, or touching the lids or lashes with instruments entering the eye.

Prolonged manipulation during the operation.

From the theatre equipment Contaminated irrigating fluid.

Faulty steriliser.

Faulty sterilising technique.

Contamination of sterile trolleys e.g. from flies.

Contamination or inactivation of disinfectant solutions.

Broken or defective packaging for an IOL or

instrument which is factory sterilised.

- Referral of faulty equipment and apparatus

Ensure that defective equipment is taken out of service, clearly labeled and disinfected, and that proper validation is performed once equipment is repaired

When equipment is broken and needs repair, it is highly important to eliminate the risk of the person performing the repair becoming infected with pathogenic organisms. Besides, it must be clear to everybody that the equipment is defect which prevents that people still try to use the defective equipment, creating potential hazards to the equipment, the person using it and the test result.

You have to be certain that equipment that is returned into service after being repaired is able to perform correctly and delivers good quality and true results.

Write an SOP that provides all the procedures that must be followed when a piece of equipment is defective; from disinfection and removal from the laboratory to returning into the laboratory following proper validation. Include in this SOP at least the following:

Procedure for disinfecting the equipment, including specification of when it should be disinfected.

Procedure ensuring that equipment is clearly labeled as defective directly after detecting that it is broken, preventing further use of the equipment. This label must also indicate if the equipment has been disinfected.

Procedure for ensuring appropriate packaging when the equipment needs to be transported, preventing further damage to the equipment.

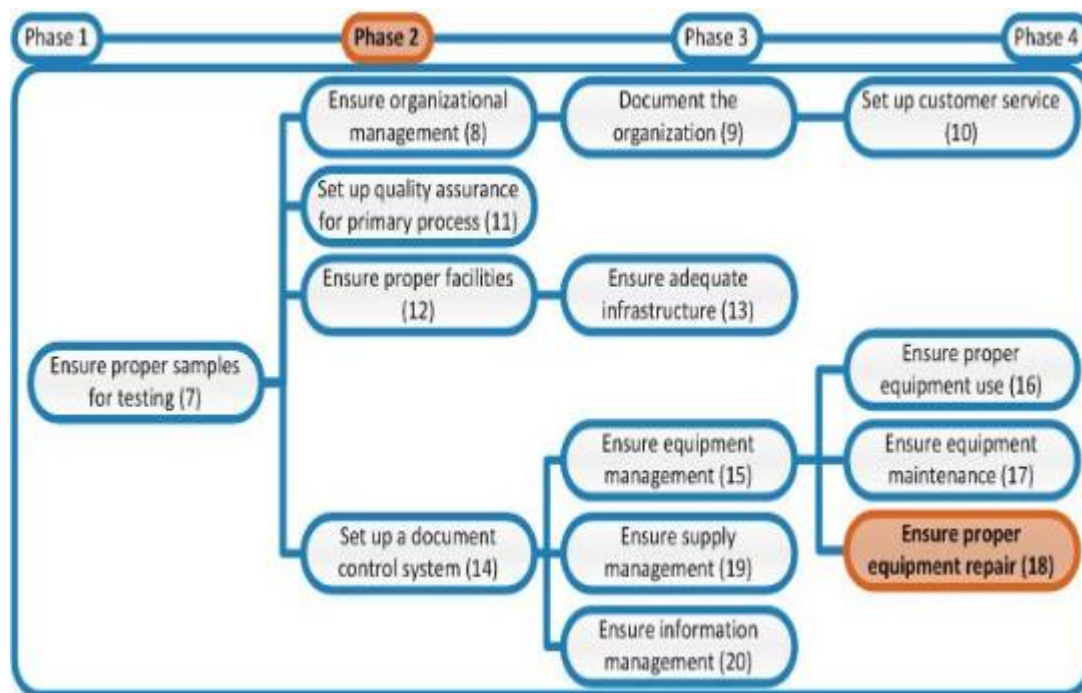
Procedure for ensuring that adequate repair is done according to the SOPs in which the procedures for maintenance of the broken piece of equipment are described (including recording of repairs in the equipment files).

Procedure that has to be followed when equipment is returned into service again after reparation: verification that the equipment is able to achieve the predefined performance requirements and that it is able to deliver a true and high quality result (which is called validation of equipment).

In the right-hand column you can find background information regarding equipment repair and maintenance from the WHO Laboratory Quality Management System (LQMS) handbook.

With regard to the last point: As the performance requirements differ per piece of equipment you cannot define these in the SOP. Therefore, include a form for recording that validation has happened. This form is similar to the equipment reception checklist (attached to the SOP for Procurement and Reception of Equipment): it is used to verify that the piece of equipment is able to achieve performance requirements formulated by you. Hence the form must have space allocated for formulation of performance requirements to which the piece of equipment must be tested and must comply with. Make sure the form guides the person validating the equipment through the whole process in a stepwise manner.

After finishing the validation the person should sign the form and give the form to the Equipment Officer for archiving in the equipment file. This form will be an appendix of the SOP.



- Monitoring and evaluation in Operation Theatre

Hospital-centric Metrics There are several performance measures of ORs, which are of special interest to the hospital involved due to their impact on productivity or revenue. Since these measures do not take into consideration the experience of patients who have undergone surgeries, we collectively define them as hospital-centric metrics. Some of the commonly cited hospital-metric metrics include OR utilisation, throughput, waiting time of surgeons, overtime costs, contribution margin, and makespan as defined and illustrated the bellow

Metric Description

OR utilization Ratio of time spent* by patients in OR to total OR time available

Throughput Number of surgical cases per unit time

Waiting time of surgeons Length of time spent by surgeons waiting prior to the start of their procedure.

Overtime costs Additional costs incurred due to performance of surgeries beyond the standard operating hours

Contribution margin is typically computed in terms of dollar per unit

OR time and is the revenue generated by a surgical case

Learning outcome 4: Decontamination in Operation Theatre instrument and apparatus

Introduction to the learning outcome

Upon completion of this unit describes the knowledge, skills and attitudes required to Decontamination in Operation Theatre instrument and apparatuses-arrangement of theatre room, Referral of faulty equipment and apparatus and Documentation of Operation Theatre technical service

Performance Standard

1. Used Instruments and equipment are decontaminated according to WHO standards
2. Theatre room is rearranged in accordance with work place set- up
3. Faulty apparatuses and instruments are referred to maintenance department according to work place policy
4. Technical services are documented in accordance with work place procedures
5. Waste is disposed with due regard to environment protection regulations

Definition of terms

Standards of Practice for the Decontamination of Surgical Instruments

Introduction

The following Standards of Practice were researched and authored by the AST Education

and Professional Standards Committee and have been approved by the AST Board of Directors. They are effective April 16, 2009.

AST developed the Standards of Practice to support healthcare facilities in the reinforcement of best practices related to the decontamination of surgical instruments in the perioperative setting. The purpose of the Standards is to provide an outline that the Certified Surgical Technologist (CST) and Certified First Assistant (CSFA) can use to develop and implement policies and procedures for the decontamination of surgical instruments. The Standards are presented with the understanding that it is the responsibility of the healthcare facility to develop, approve and establish policies and procedures for cleaning and disinfecting surgical instruments according to established healthcare facility protocols.

Rationale

The following are Standards of Practice related to the proper decontamination of surgical instruments (henceforth, simply referred to as instruments) in the perioperative setting. Instruments that are opened on the sterile field, whether used or not used, during the surgical procedure must be thoroughly decontaminated prior to disinfection and/or sterilization. The terms cleaning and decontamination are often used synonymously to indicate a physical and chemical process of removal of organic material, soil and debris, and microorganisms from inanimate objects, such as instruments. The term cleaning frequently, as indicated in the prior sentence, refers to the removal of microorganisms as opposed to killing. However, the Occupational Safety and Health Administration defines

decontamination as, “the use of physical or chemical means to remove, inactivate, or destroy blood-borne pathogens on a surface or item to the point where they are no longer capable of transmitting infectious particles and the surface or item is rendered safe for handling, use, or disposal.”²⁷ For the remainder of this document, the terms cleaning and decontamination will be used synonymously, and the OSHA definition will apply to both terms, meaning the process removes, inactivates or destroys microorganisms.

When performed properly, cleaning effectively reduces the bioburden in order to prepare the instruments for disinfection and sterilization. The importance of this step cannot be overemphasized since organic material, soil, and debris can block the disinfectant or sterilizing agent from making complete contact with the surface of the instruments. Additionally, cleaning allows for the safe handling of the instruments by healthcare workers (HCWs).

Standard of Practice I

The cleaning of instruments should begin during the surgical procedure to prevent drying of blood, soil and debris on the surface and within lumens.

1. The CST in the first scrub role should keep the instruments free of debris and blood during the surgical procedure.
 - A. The instruments should be wiped clean using a sterile, water-moistened sponge. Care must be taken that the sponge is not used on the tissues of the patient.
 - B. Instruments with lumens should be flushed with a sterile, water-filled syringe to remove blood and debris and prevent drying of the gross soil.

C. Instruments that may not be used for the remainder of procedure, eg, acetabular reamers used during a total hip arthroplasty, may be placed into a basin containing sterile water to soak.

D. Saline must not be used, since the chloride ions can cause pitting and deterioration of the finish on the surface of the instruments.

Standard of Practice II

The cleaning of instruments should continue at the point of use post-procedure, including sorting and disassembly of instruments, containment and transportation the decontamination room.

1. Post-procedure, after removal of the sterile gown and gloves, the CST should wear personal protective equipment (PPE) when breaking down the sterile back table.

2. Instruments should never be soaked in saline or sodium hypochlorite (bleach) solution. The chloride ions in both solutions are highly corrosive, causing the breakdown of the finish on instruments, as well as the metal.

A. It is recommended that soaking soiled instruments begin in the OR at the completion of the procedure. The instruments can be placed in a basin containing a mixture of sterile water and enzymatic detergent. Refer to the manufacturer's instructions for the correct amounts of sterile water and enzymatic detergent to be mixed.

3. All instruments that were on the sterile field, whether used or not used, are

considered contaminated, and a possible source of microorganisms that could cause an infection in HCWs and patients.

A. All instruments must be properly separated and placed in leakproof, puncture-resistant containers that are marked with a biohazard label to allow easy identification by other HCWs that the contents are contaminated and therefore hazardous.

(1) The infection hazard to the surgical team members is greatest during the handling and separation of contaminated instruments at the point of use. However, separation is best done in the OR by the CST who is familiar with the back table set up, including location of the sharps container, as well as other sharps such as trocars in order to prevent injury to other surgical team members. (2) A sponge moistened with water should be used to wipe gross soil and blood from the instruments that were used during the surgical procedure. It is recommended not to clean the instruments within a water-filled basin in order to prevent splashing of the fluid on the floor or other surfaces of the OR

(3) Ringed instruments should be placed in a carrier arranged in single layers or placed on stringers. Box locks should be opened. Assembled instruments should be taken apart. Heavy instruments and delicate instruments, such as microsurgical instruments should be placed in separate trays.

(4) Reusable sharp instruments with points or edges should be placed in a separate puncture-resistant, closable container.

(5) Instruments should be kept wet in the transport container by covering with a water-soaked towel or spray, foam or gel product specifically intended for this purpose. The instruments should not be transported in the water and enzymatic soaking solution in order to prevent splashing and reduce the weight of the transport container.

The enzymatic solution should be properly discarded according to surgery department policy by the CST who is wearing PPE.

B. Contaminated instruments should be contained during transport from the point of use to the decontamination area.

(1) Contaminated instruments should be handled as little as possible at the point of use and should be immediately contained and transported to the decontamination area. Immediate containment

and transport reduces the risk of surgical personnel's contact with the contaminated instruments.

(2) Containment should be achieved through the use of some type of container that has been identified to prevent surgical personnel and other HCWs from contact with the contaminated instruments and prevention of airborne microorganisms during transport.

The type of container to be used depends on the items to be transported. Types of recommended containers include closed carts, bins with lids, rigid sterilization container systems, and impermeable bags that can be used alone or in combination.

(3) Rigid sterilization container systems with intact, dry filters and closed valves are an accepted method for the transportation of contaminated instruments. However, the use of the container for

transportation purposes should be confirmed by consulting the manufacturer's instructions. Some manufacturers recommend not using the same container for transporting contaminated instruments that is also used for the sterilization of the instruments.

(4) The rigid sterilization container system requires no additional covering, unless the external surfaces have been contaminated with blood and/or body fluids. If the container has been handled

and touched by a person who has had contact with blood or body fluids, such as the CST, it should be assumed it is contaminated. In this instance, the container should be enclosed in a red

biohazard bag, bin with a lid or closed cart.

When purchasing a rigid sterilization container system, the healthcare facility should

consult the manufacturer's information to confirm if the container can be easily and effectively decontaminated.

(5) The external surfaces of bins with lids, closed carts and other containers should be decontaminated after each use with an Environmental Protection Agency (EPA) - registered,

intermediate-level disinfectant. The containers should be thoroughly wiped down externally and internally. The use of a cart wash system is recommended for decontaminating closed carts. Additionally, routine cleaning of the case cart wheels should be performed to remove string and other debris to maintain the easy movement of the wheels.

Standard of Practice III

Cleaning/detergent agents should be selected that will not damage the cleaning equipment and effectively clean instruments.

1. The chemicals in the cleaning agent(s) should not be corrosive to the cleaning equipment including the ultrasonic cleaner, washer-decontaminator, or washer-sterilizer.

2. The chemicals in the cleaning agent(s) should not cause electrolytic action between the instruments and cleaning equipment.

3. The cleaning agent(s) should not be corrosive and damaging to the instruments.

4. Cleaning agent(s) should be easily removed from the instruments by rinsing with water in order to avoid residual chemicals that could corrode and damage the instruments as well as present a danger to the patient.

5. It is recommended to select a detergent with a pH between 7-10 to use in the cleaning of instruments.¹⁶ Detergents that have a pH of higher than 7 are more effective in the removal of organic debris such as blood, fat and feces.

6. Antimicrobial solutions, such as iodophors that are used for skin antisepsis, should never be used for cleaning instruments.

7. Manufacturer's instructions for the use of detergents should always be followed.

The instructions should be kept on file and accessible at all times by HCWs.

8. The following are the ideal characteristics of a cleaning agent.

- A. Low sudsing/foaming
- B. Easily rinsed off
- C. Disperse organic soil
- D. Biodegradable
- E. Nontoxic
- F. Nonabrasive
- G. Effective on all types of organic soil
- H. Cost-effective
- I. Long shelf life

Standard of Practice IV

Cleaning may be performed manually, mechanically or a combination of both. The

selection of the cleaning method should be based upon the type of device and manufacturer's recommendations. However, cleaning alone may not be sufficient to decontaminate items that present a high risk of disease transmission such as surgical instruments and therefore, should undergo a microbicidal process.

1. The manufacturer should provide written instructions for the reprocessing of the instrument(s) or device(s) that include recommendations for the type of cleaning method to be used, type of cleaning equipment and cleaning agent(s).

A. The cleaning method should not affect the function of the instruments or

devices and should be safe to use by HCWs.

B. The manufacturer's instructions should be kept on file and accessible at all times by HCWs.

C. Prior to HCWs using cleaning equipment and/or cleaning agents that are not recommended by the manufacturer for particular instruments or devices, the HCW should

contact the manufacturer for recommendations, as well as contact the manufacturers of the cleaning equipment and cleaning agent(s).

2. Manual cleaning is recommended for delicate instruments and devices, such as microsurgical instruments, lensed instruments, power equipment, and other instruments that cannot tolerate an automated cleaning process.

A. Immersible instruments and devices should be kept submerged in the water to prevent aerosolization. Non-immersible instruments and devices should be cleaned according to the manufacturer's instructions.

B. The water with detergent should be kept at a temperature in a range of 27°

C to 44° C (80° F – 110° F). Temperatures over 110° F cause coagulation and thus prevent removal of protein substances. Water temperatures that are too cold may not activate the detergent.

C. After cleaning, instruments and devices should be thoroughly rinsed to remove detergent residue and debris.

(1) Rinsing is extremely important, because residuals can reduce the efficacy of the disinfection and sterilization processes and possibly cause damage to the tissues of the surgical patient.

D. Scouring pads and abrasive cleaning agents should not be used for cleaning instruments and devices in order to prevent damage to the items.

E. Only brushes designated for use in cleaning instruments and devices should be purchased by the healthcare facility.

(1) Reusable brushes create a risk for cross-contamination. Reusable brushes should be cleaned and decontaminated at least daily or when heavily soiled. Brushes that show wear should be discarded.

(2) Brushes used to clean instruments and devices with lumens must be the correct size. If the brush is too large, it will not properly fit into the lumen; if shoved into the lumen it could damage the

instrument or device and possibly become stuck within. If too small, the brush will not make complete contact with the lumen's surface and prevent thorough cleaning.

F. A three-sink method should be used for manual cleaning: A sink with tap water and detergent; second sink with tap water for rinsing; third sink with distilled/de-ionized water as a second rinse to aid in preventing staining.

3. The use of mechanical cleaning equipment is recommended as the primary method for decontamination of instruments and devices that can withstand the process. Mechanical cleaning equipment provides the advantage of exposing the instruments and devices to a microbicidal process.

A. The use of mechanical cleaning equipment significantly reduces the amount of contact time between the HCW and contaminated items.

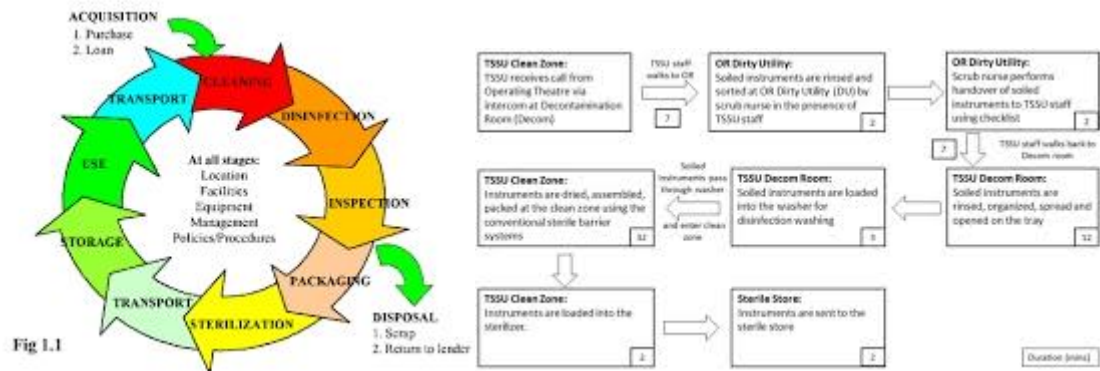
B. Thermal disinfection can be accomplished with the use of washer-sanitizers, washer-decontaminators, washer-disinfectors, and washer-sterilizers. Washer sanitizers provide the lowest level of disinfection, and washer-sterilizers offer the highest level of disinfection.

(1) Thermal disinfection should only be used to render items safe to handle by HCWs not wearing PPE and not as a process in which they are ready for reuse in surgery. Critical items should always be put through a sterilization process as well as semi-critical items should be either sterilized or high level disinfected.

(2) To ensure proper functioning of the equipment, routine maintenance should be performed according to the manufacturer's instructions, the strainer should be cleaned on a daily basis, and the operating instructions provided by the manufacturer should be followed.

(3) Time and temperature should be monitored and documented.

(4) HCWs must be careful when removing hot items from the equipment in order to avoid burns. The items may be wet with hot water. Additionally, water that drips on the floor may make it slippery; the water should be wiped from the floor.



- Re-arrangement of theatre room



he circulator should check the operating room schedule the day before surgery for any changes that may have been made. The operating room specialist should be in the assigned room in plenty of time to prepare for the surgical procedure.

Centralized Materiel Service should be notified of any special equipment and/or instruments that may be needed for the case that are not readily available.

b. Personal cleanliness is extremely important for the operating room specialist.

Before performing the duties described in the following paragraphs for preparation of the operating room, the circulator should wash his hands thoroughly and don a surgical hat. The surgical cap must cover the hair completely to prevent possible contamination of the sterile area by falling hair or dandruff. He should then don a clean, cotton scrub suit before entering the semi-restricted areas of the surgical suite.

Before entering a restricted area, the OR specialist must don a surgical mask per local SOP. The mask protects the patient from bacteria exhaled by operating room personnel. The mask must fit snugly around the nose and mouth to filter the air through it rather than

around the sides of the mask. The mask should be changed whenever it becomes damp and after each procedure.

The correct procedures for donning scrub suit, surgical hat, shoe covers, and surgical mask are provided in detail in Subcourse MD0933, Scrub, Gown, and Glove Procedures, which you should have studied prior to this subcourse. Now you are ready to prepare the operating room for surgery.

c. Damp dust the operating room unless this has already been done by personnel on the previous shift.

Concurrent with dusting, check equipment, arrange furniture, and restock supplies. After damp dusting with a cloth soaked in disinfectant solution prescribed by local policy, wet vacuum the floor using a disinfectant prescribed by local policy. Dry dusting and mopping is never done in the operating room because it raises dust that contains bacteria.

(1) Damp dusting should be done before the first scheduled incision time of the day.

(2) Establish and follow a definite order when damp dusting furniture. Start with the tallest equipment and work down since this method helps the settling of airborne microorganisms. Damp dust the operating room overhead light first, then the operating table. Work from the center of the room to the perimeter (outer limits) and from the tallest item to the lowest. If you are called from the room, leave the damp dusting cloth on the item being dusted; this will serve as the starting point when resuming dusting duties.

(3) As the damp dusting is accomplished, set up the equipment and check each item for proper functioning. This will save time and energy. This includes such things as:

(a) Switch on the overhead light to ensure proper functioning.

(b) Check the operating table for proper working order.

(c) Check the suction machine, the electrosurgical unit, and other pieces of equipment in the operating room whether or not they are to be used.

(d) Line the kick buckets with plastic bags.

(e) Check the supply cabinets for stock. Restock, if necessary.

(4) Consult with the scrub on the arrangement of the furniture for the surgical procedure. In general, the area chosen for the sterile setup should be away from doorways and traffic. It should be in the most closed-in area away from the cabinets that are to be opened during the setup. As damp dusting continues, arrange the furniture for the sterile setup so that the scrub can work within the sterile field and the circulator can work outside the field.

d. Check with the scrub for any special equipment he may need for the case.

e. If the instruments were not wrapped and sterilized on the preceding shift, ensure that instrument sets are placed in the autoclave.

f. Check that all equipment needed to position the patient is in the room. If not, get the necessary equipment.

g. Discuss with the scrub the sterile supplies needed for the case (be sure to discuss the kind and amount of sutures needed) and then bring the supplies into the room. The surgeon's preference card will list the types and sizes of sutures needed for the procedure.

h. Place sterile goods on the tables or stands where they will be used to avoid having to move them from one place to another. You should place the various sterile items as indicated in (1) through (6) below.

(1) The linen or drape pack on the large instrument table (backtable).

(2) The gown pack on the prep table.

(3) The Mayo tray on the Mayo stand.

(4) The sterile basin set into the ring stand.

(5) The prep set placed on the prep table. Only after the scrub dons the gown and the wrapper is removed,

(6) The instrument set on a ring stand or table.

i. Put sterile packages that the scrub will not need immediately (such as suction tubing and culture tubes) on the utility table and open them after the scrub has prepared a space for them. You may also place on the utility tables supplies needed from the sterile supply cabinets such as knife blades, needles, etc.

j. Do not place sterile supplies on the operating table nor on the anesthetist's equipment (the anesthesia apparatus and the anesthetist's table) because these areas are for the use by the anesthetist only.

k. Open the sterile supplies. Before any sterile supplies are opened, however, the integrity of every package is checked for tears, punctures, watermarks, expiration date, and the sterilization indicator. Tears, punctures, and watermarks on a sterile pack indicate that the supplies are unsafe to use. The sterilization indicator shows whether the pack has been through the sterilization process. The date of expiration will tell you whether it is too old for safe use. If the package is in any way compromised, it must be discarded and a new pack secured.

l. Open the packs and sets in the order in which the scrub will need them. Open the sterile gown first; the scrub will need this immediately upon entering the room.

(1) Remove the tape from the packages, unwrap the item, and check the indicator tape to ensure item has been through the sterilization process. If the indicator tape shows no or incomplete sterilization process, discard the package and secure a new one.

(2) After opening the pack containing the scrub's gown, open the basin set, the linen pack, the prep set, and the instrument set.

m. All sterile wrappers are to be removed in the same general manner. Open the wrapper so that your hand and arm do not pass over any part of the inside of the wrapper that has been exposed

- Documentation of Operation Theatre technical service

An operating theatre book is essential: it contains detailed records of the patients operated on and the surgical procedures performed. This is useful information for planning and ordering supplies. All information in the operating theatre book is considered to be confidential.

Information to be recorded in the operating theatre book
Consecutive patient number (OT number)

Patient name

Age/sex

Patient's admission number

Diagnosis

Aetiology

Surgical procedure done

Surgeon

Type of anaesthesia given

Anaesthetist

For ease of identification, each patient's first operation should be entered with a red pen and subsequent operations in blue/black.

For accurate record-keeping there are two or three numbers recorded in the operation book for each operation:

- the number that corresponds to the number of operations performed in the operation theatre (the consecutive OT number)
- the patient's admission number, which corresponds to the number written on the individual patient record

Recording the diagnosis/aetiology may only be necessary for the initial operation (see Fig. 39).

Other useful documents and checklists include:

- operation lists
- compress check for laparotomies
- weekly maintenance checks
- weekly oiling of instruments
- duty roster
- diary of ward rounds for compilation of operating lists

The operating theatre book.

The first two patients are routine cases and their entries are written in black; note that in the “diagnosis” column the diagnosis is replaced by their “OT” number which corresponds to the operation number of their initial surgical intervention: this saves time.

Subsequent patients are new admissions (admitted during an influx of wounded); their entries are written in red so that they can be easily identified as new cases.

Note that the triage numbers are also written in the patient number column. This is because very often patients admitted during an influx are not allocated an admission number before being transferred to the operating theatre, but it is nevertheless essential to be able to identify each patient accurately.

DATE	OPERATION NUMBER	NAME	AGE	SEX	PATIENT NUMBER	DIAGNOSIS / OT no.	WET/DRY	OPERATION PROCEDURE	ANESTHETIST	ANESTHESIA	WOUNDED
6/7	3411	Salamet Khan	11	M	1863	2896		Skin graft @ arm	KC	K	GW
6/7	3412	Lal Mohammed	26	M	1876	3327		hPC @ leg	KC	K	GW
6/7	3413	Mohammed Kham	52	M	1917 007	S.I. chest, pneumonia		hPC @ arm & wound excision	KC	K	GW
11	3414	Belqam		F	1919 009	S.I. abdomen		Laparotomy, colostomy	KC	GA	GW
11	3415	Sada Rahim	8	M	1913 002	S.I. @ leg		wound excision	PP	K	JV
11	3416	Mahmud	7	F	1915 002	S.I. @ leg		wound excision	PP	K	JV
11	3417	Sunayna	2	F	1912 006	S.I. both arms		wound excision	PP	K	JV
11	3418	Shirika	11	F	1911 006	S.I. @ leg @ arm		wound excision	PP	K	JV
11	3419	Fahima	5	F	1918 008	S.I. skull		wound excision	KC	K	GW
11	3420	Mohammed Jan	60	M	1916 005	S.I. @ arm		wound excision	KC	K	GW

SHORT RESPONSES FOR LEARNING OUTCOME 4

1. List useful documents and checklists include:
2. As the damp dusting is accomplished, set up list the item you will check for for proper functioning
3. After cleaning, instruments and devices should be thoroughly rinsed explain the importance of rinsing,
4. The CST in the first scrub role should keep the instruments free of debris and blood during the surgical procedure. explain how this is done

SHORT RESPONSES ANSWERS FOR LEARNING OUTCOME 4

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2. As the damp dusting is accomplished, set up list the item you will check for for proper functioning

(a) Switch on the overhead light to ensure proper functioning.

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3. After cleaning, instruments and devices should be thoroughly rinsed explain the importance of rinsing,

Rinsing is extremely important, because residuals can reduce the efficacy of the disinfection and sterilization processes and possibly cause damage to the tissues of the surgical patient.

(d) Line the kick buckets with plastic bags.

(e) Check the supply cabinets for stock. Restock, if necessary.

4. The CST in the first scrub role should keep the instruments free of debris and blood during the surgical procedure. explain how this is done

A. The instruments should be wiped clean using a sterile, water-moistened sponge. Care must be taken that the sponge is not used on the tissues of the patient.

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